NEET-SS / FET / INI-SS HIGH-YIELD NOTES

Oncology

1. bone-forming tumours
2. cartilage tumours
3. giant cell and cystic lesions
4. small round cell tumours
5. soft-tissue sarcoma
6. metastatic bone disease

Bone-forming tumours

1. Overview and general principles

- **Bone-forming (osteogenic) tumours** are defined by tumour cells that **produce osteoid or bone directly**.
- **WHO 2020 classification of osteogenic tumours:**
  - **Benign:** osteoma, osteoid osteoma.
  - **Intermediate (locally aggressive):** osteoblastoma.
  - **Malignant:** osteosarcoma and its subtypes (conventional, telangiectatic, small cell, low-grade central, parosteal, periosteal, high-grade surface, secondary).
- **Osteosarcoma is the commonest primary malignant bone tumour** (excluding myeloma, which is a marrow neoplasm). Chondrosarcoma is second, Ewing sarcoma third (second in children).
- **Clinical age clue:** under 5 years, a destructive bone lesion is usually metastatic neuroblastoma or LCH; 10-25 years osteosarcoma or Ewing; over 40 years metastasis or myeloma first, then chondrosarcoma.
- **Investigation sequence for a suspicious lesion:** plain radiographs in two planes, **MRI of the whole bone** (extent, soft tissue mass, skip lesions, neurovascular relations), **CT chest** (lung metastases), bone scan or **PET-CT** (bone metastases), then **biopsy last**, at the treating sarcoma centre.
- **Biopsy rules (Mankin):** core needle preferred; tract **in line with the definitive incision** so it can be excised en bloc; go **through a single muscle compartment**, not between intermuscular planes; avoid neurovascular bundles and joints; meticulous haemostasis; drains brought out in line with the incision. Mankin showed that biopsies done outside specialist centres caused significantly more errors and changes in treatment, including unnecessary amputation.

Surgical margins (Enneking)

Margin	Plane of dissection	Result for high-grade sarcoma
Intralesional	Through the tumour	Gross residual tumour; nearly always recurs

Margin	Plane of dissection	Result for high-grade sarcoma
Marginal	Through the reactive zone (pseudocapsule)	Leaves satellite lesions; high recurrence
Wide	Through normal tissue with a cuff around the reactive zone	Standard goal for limb salvage
Radical	Entire anatomical compartment removed	Rarely needed now with chemotherapy

Enneking (MSTS) staging

Stage	Meaning
Benign 1 (latent)	Inactive, well-marginated, often heals spontaneously (e.g. NOF)
Benign 2 (active)	Growing, contained within bone (e.g. most osteoid osteomas)
Benign 3 (aggressive)	Breaches cortex, may metastasise rarely (e.g. osteoblastoma, GCT)
IA / IB	Low grade, intracompartmental (A) or extracompartmental (B)
IIA / IIB	High grade, intra- or extracompartmental. **Most osteosarcomas are IIB**
III	Any grade with distant metastases

- **AJCC 8th edition (appendicular skeleton):** T1 is 8 cm or less, T2 over 8 cm, T3 discontinuous tumours (skip lesions) in the primary bone; M1a lung only, M1b other distant sites. Stage IV = metastatic (IVA lung only; IVB other sites or nodes).

2. Osteoma and bone island

- **Osteoma:** dense lamellar bone on the surface of the **skull, facial bones and paranasal sinuses** (frontal sinus commonest). Usually asymptomatic.
- **Gardner syndrome:** multiple osteomas + **familial adenomatous polyposis** (APC gene, chromosome 5q) + desmoid tumours and epidermoid cysts. Colonic polyps have near 100% malignant potential, so osteomas of the jaw should prompt colonoscopy.

- **Bone island (enostosis):** intramedullary compact bone; **"thorny" radiating spicules (brush border)** blending into surrounding trabeculae; commonly pelvis and proximal femur.
  - Usually cold or mildly warm on bone scan, which helps distinguish it from **osteoblastic metastasis** (hot).
  - Over 2 cm = **giant bone island**. Growth over time warrants biopsy only if behaviour is atypical.
- **Osteopoikilosis:** multiple bone islands around joints (LEMD3 gene), asymptomatic; **Buschke-Ollendorff** when combined with skin lesions.

3. Osteoid osteoma

Epidemiology and pathology

- About 10-12% of benign bone tumours. **Age 5-25 years**; **male to female about 2-3:1**.
- **Sites: proximal femur (femoral neck and intertrochanteric region) is the commonest**, then tibia; about 10% in the spine (**posterior elements**, lumbar commonest); hand and foot.
- **Nidus under 1.5 cm** (some texts 2 cm): interlacing osteoid trabeculae lined by osteoblasts in vascular fibrous stroma, surrounded by **reactive sclerosis**. Not malignant; no malignant transformation.
- Pain mediated by **prostaglandin E2 and COX-2** production in the nidus; the nidus contains **unmyelinated nerve fibres**.
- **FOS gene rearrangements** are now recognised in both osteoid osteoma and osteoblastoma.

Clinical features

- **Night pain relieved dramatically by aspirin or NSAIDs** (classic; about 75%).
- **Spine: painful scoliosis** (stiff, non-structural), the **lesion is on the concave side at the apex** of the curve; paravertebral spasm. Early removal (within about 15 months) allows the curve to resolve.
- **Intra-articular (hip, elbow, ankle):** synovitis, effusion, stiffness, little sclerosis; mimics inflammatory arthritis or infection; delayed diagnosis common.

- **Near the physis in children:** overgrowth and limb length discrepancy; muscle wasting.

Imaging

- **Radiograph:** small **radiolucent nidus with dense surrounding sclerosis** and cortical thickening (cortical type commonest); the nidus may be hidden by sclerosis.
- **CT (thin slices) is the investigation of choice to locate the nidus**; nidus may show central mineralisation (target).
- **Bone scan:** intense uptake, **"double-density sign"** (hot nidus within a less hot reactive zone).
- **MRI:** marked marrow and soft tissue oedema may mislead (mimics infection or tumour); dynamic contrast shows early arterial enhancement of the nidus.

Differential diagnosis

- **Brodie abscess** (serpiginous lucent tract, often crosses physis, no nidus calcification), **stress fracture** (linear lucency perpendicular to cortex), osteoblastoma (larger), chronic osteomyelitis, intracortical haemangioma.

Management

Option	Indication / notes
NSAIDs	Many lesions burn out in 2-4 years (natural history); acceptable if tolerated
CT-guided radiofrequency ablation (RFA)	**Treatment of choice** for most accessible lesions; primary success about 90%, recurrence about 5-10%, can be repeated. Heats nidus to about 90 degrees C for 4-6 min
Surgical excision (en bloc or burr-down)	Lesion near skin (burn risk), near nerve roots or spinal cord (relative caution with RFA), failed RFA, need for histology
Spine	Excision or RFA with neural protection; early treatment to prevent structural scoliosis

- Recurrence of pain after treatment means incomplete nidus removal.

> Exam pearl: Osteoid osteoma = nidus under 1.5 cm, night pain relieved by NSAIDs (PGE2/COX-2), proximal femur commonest, CT best to find the nidus, "double-density" on bone scan, painful scoliosis with the lesion on the CONCAVE side at the apex, RFA is the treatment of choice.

4. Osteoblastoma

- **Over 1.5-2 cm**, histologically similar to osteoid osteoma but **progressively growing** and **locally aggressive** (Enneking benign 3 in many cases).
- **Age 10-30**, male predominance.
- **Spine and sacrum are the commonest sites (about 40%)**, again in the **posterior elements**; also long bones (metadiaphysis) and talus.
- Dull ache, **less night pain and poor response to NSAIDs**; may cause **radiculopathy or cord compression**, and painful scoliosis.
- **Imaging:** expansile lytic lesion with variable mineralisation and a thin rim of reactive bone; **secondary ABC change** is common; soft tissue extension.
- **Differential: osteoblastoma-like osteosarcoma** (permeative margin, infiltration of host trabeculae), ABC, GCT.
- **Treatment: extended curettage with adjuvant (high-speed burr, phenol, cryotherapy) and grafting**, or **en bloc resection** for aggressive and spinal lesions (with stabilisation). Recurrence after curettage about 10-20%. Radiotherapy is avoided.
- **Epithelioid ("aggressive") osteoblastoma** has a higher recurrence risk.

5. Conventional osteosarcoma

Epidemiology and risk factors

- **Bimodal:** peak in **adolescence (10-20 years, growth spurt)** and a second peak **over 60 years** (secondary: Paget disease, irradiation). Slight male predominance.
- **Sites:** metaphysis of long bones with the fastest growth: **distal femur > proximal tibia > proximal humerus** (about 50% around the knee). Axial (pelvis, spine) lesions in older patients carry a worse prognosis.
- **Genetic associations:**

  - **Hereditary retinoblastoma (RB1, chromosome 13q14)**: large increase in risk, also in irradiated fields.
  - **Li-Fraumeni syndrome (TP53, 17p13)**: osteosarcoma, breast cancer, brain tumours, leukaemia, adrenocortical carcinoma.
  - **Rothmund-Thomson syndrome (RECQL4)**: poikiloderma, radial ray defects, osteosarcoma.
  - Werner (WRN) and Bloom (BLM) syndromes (RecQ helicases).
- **Secondary osteosarcoma:**
  - **Paget sarcoma:** in about 1% of Paget disease (higher in polyostotic disease); pelvis, femur, humerus; new pain or swelling in pagetic bone; **very poor prognosis** (5-year survival under 10-15%).
  - **Post-radiation sarcoma (Cahan criteria):** arises in the radiation field, latency of **over 3-4 years** (median about 10-15 years), histologically different from the original lesion. Osteosarcoma is the commonest radiation-induced bone sarcoma.
  - Others: fibrous dysplasia, bone infarct, chronic osteomyelitis (rare).

Pathology

- **Diagnostic criterion: malignant spindle (pleomorphic) cells producing osteoid directly**. Amount of osteoid can be small.
- High grade; histological subtypes of conventional OS: **osteoblastic (commonest), chondroblastic, fibroblastic**; subtype does not change treatment.
- Grows through the cortex and lifts periosteum; spreads within the medulla (**skip metastases** in about 5-10% of patients, i.e. discontinuous foci in the same or transarticular bone) and **haematogenously** (lymph node spread rare).
- **Lungs are the commonest site of metastasis** (about 80-90% of metastatic cases), then bone. About 15-20% have detectable metastases at presentation, but about 80% have micrometastases, which is why surgery alone gave only 10-20% survival before chemotherapy.

Clinical features

- **Pain** (initially activity-related, then constant and at night), swelling, warmth, restricted joint movement; pathological fracture in about 5-10%.

- Often attributed to sports injury, causing diagnostic delay.
- **Serum alkaline phosphatase and LDH:** raised in about half; **high LDH or ALP indicates worse prognosis**, and ALP is used to monitor relapse.

Radiology

- **Metaphyseal, poorly defined, permeative, mixed lytic-sclerotic** lesion with a wide zone of transition.
- **Cloud-like osteoid matrix** in the soft tissue mass.
- **Periosteal reactions: Codman triangle** (elevated periosteum at the edge; seen in any aggressive lesion, not specific), **sunburst** (spiculated radiating) pattern, interrupted lamellar reaction.
- **MRI:** extent of marrow involvement and **skip lesions** (image the **whole bone and adjacent joint**), physeal and epiphyseal involvement, joint invasion, neurovascular encasement - all decide whether limb salvage is possible.
- **CT chest** for lung nodules; bone scintigraphy or **PET-CT** for skeletal metastases.

Prognostic factors

Factor	Unfavourable
Metastases at diagnosis	**Most important adverse factor** (5-year survival about 20-30%; better if lung only and resectable)
Histological response to neoadjuvant chemotherapy (Huvos)	**Less than 90% tumour necrosis = poor responder**
Site	Axial / pelvic tumours
Size	Large volume tumours (T2)
Surgical margin	Inadequate margin raises local recurrence
Laboratory	Raised LDH and ALP
Skip lesions	Behave like metastatic disease

Huvos (Rosen-Huvos) grading of chemotherapy response

Grade	Necrosis	Interpretation
I	Little or no effect (under 50%)	Poor responder
II	50-89%	Poor responder
III	90-99%	**Good responder**
IV	100% (no viable tumour)	**Good responder**

Management

1. **Neoadjuvant (pre-operative) chemotherapy** for about 10 weeks: **MAP = high-dose Methotrexate (with leucovorin/folinic acid rescue), doxorubicin (Adriamycin) and cisPlatin**. This was developed from the **Rosen T-10 protocol** at Memorial Sloan Kettering.
2. **Restaging** with MRI and CT chest.
3. **Surgery with wide margins**: limb salvage in about 85-90% of patients today.
4. **Adjuvant (post-operative) chemotherapy** continuing MAP (total treatment about 29-30 weeks); the resected specimen gives the Huvos response grade.
5. **Pulmonary metastasectomy** for resectable lung metastases (at presentation or relapse) improves survival; it is a standard part of treatment.

- **Chemotherapy toxicities:** methotrexate (renal failure, mucositis; monitor levels, glucarpidase if delayed clearance), doxorubicin (**cumulative dose-related cardiomyopathy**; dexrazoxane is cardioprotective), cisplatin (**ototoxicity**, nephrotoxicity, neuropathy), ifosfamide (haemorrhagic cystitis, prevented by **mesna**; Fanconi-type nephrotoxicity).
- **Radiotherapy:** osteosarcoma is **relatively radioresistant**; used only for unresectable tumours (spine, pelvis, skull) or palliation, often as proton or carbon-ion therapy.
- **Survival:** localised extremity disease with modern treatment about **65-70% at 5 years**; metastatic about 20-30%. Survival has not improved substantially since the 1980s.

Landmark trials and evidence

- **MIOS (Multi-Institutional Osteosarcoma Study, Link et al., NEJM 1986)**: randomised trial showing that **adjuvant chemotherapy dramatically**

improves relapse-free survival (about 66% versus 17% at 2 years) compared with surgery alone. Established chemotherapy as standard.

- **Simon et al. (MSTS, JBJS 1986)**: **limb salvage for distal femoral osteosarcoma gave survival equal to amputation** when wide margins were obtained.
- **EURAMOS-1 (published 2015-2016)**: largest osteosarcoma trial. In **poor responders**, adding **ifosfamide and etoposide (MAPIE)** to MAP **did not improve event-free survival** and increased toxicity. In **good responders**, adding **pegylated interferon alfa-2b** to MAP gave no significant benefit. Conclusion: **MAP remains the standard for all**, and post-operative chemotherapy should not be changed on the basis of poor response.
- **INT-0133 (Children's Oncology Group)**: **mifamurtide (MTP-PE, liposomal muramyl tripeptide)**, an immune macrophage activator, improved overall survival when added to chemotherapy (results debated); **approved in Europe, not by the US FDA**.
- Targeted therapy: multi-kinase inhibitors (regorafenib, cabozantinib) show modest activity in relapsed disease only.

Surgical options

- **Limb salvage (contraindications are relative):** major neurovascular encasement that cannot be reconstructed, extensive soft tissue contamination (e.g. poorly placed biopsy, displaced pathological fracture through tumour), infection, very young child with large predicted leg-length discrepancy (now often managed with growing prostheses).
- **Reconstruction after resection:**
  - **Modular endoprosthesis (megaprosthesis)**: commonest; rotating-hinge knee for distal femur or proximal tibia.
  - **Expandable (growing) prosthesis** in skeletally immature children: non-invasive magnetic lengthening types avoid repeated surgery.
  - **Allograft** (osteoarticular or intercalary), **allograft-prosthesis composite**, vascularised fibula, recycled autograft (irradiated, frozen in liquid nitrogen or pasteurised).
  - **Rotationplasty (Van Nes)**: for distal femoral tumours in young children; the distal leg is rotated **180 degrees** and attached to the femur so that the **ankle functions as a knee** (plantarflexion = knee extension) and the

foot fits a below-knee type prosthesis. Durable, active, no prosthetic loosening; cosmetic appearance is the main drawback.
    - **Amputation:** above-knee or hip disarticulation when limb salvage cannot achieve wide margins; same survival as limb salvage.
- **Proximal tibial resection:** the **extensor mechanism must be reconstructed** (medial gastrocnemius rotational flap with patellar tendon reattachment), and the gastrocnemius flap also covers the implant, reducing infection.
- **Endoprosthetic failure (Henderson classification):** type 1 **soft tissue failure** (instability, tendon failure), type 2 **aseptic loosening**, type 3 **structural failure** (implant or periprosthetic fracture), type 4 **infection**, type 5 **tumour progression**. Infection is the most feared and is particularly common after proximal tibial and pelvic reconstruction.
- **Local recurrence** after limb salvage is about 5-10% and carries a poor prognosis.
- **Follow-up:** CT chest and limb imaging every 3 months in years 1-2, then less frequently to 10 years; monitor late effects (cardiac, renal, hearing, fertility, second malignancy).

> Exam pearl: Conventional osteosarcoma: adolescent distal femoral metaphysis, mixed lytic-sclerotic with Codman triangle and sunburst. MAP neoadjuvant chemotherapy, wide resection, adjuvant MAP. Most important prognostic factor = metastases at diagnosis; next = Huvos necrosis (90% or more = good responder). EURAMOS-1: intensifying chemotherapy for poor responders does not help.

6. Subtypes of osteosarcoma

Subtype	Key features	Treatment and prognosis
Telangiectatic	Purely lytic, expansile, haemorrhagic; **fluid-fluid levels on MRI mimic ABC**; malignant cells in the septa (thick, nodular, enhancing septa on MRI)	As conventional (chemotherapy + wide resection); prognosis similar to conventional; pathological fracture common

Subtype	Key features	Treatment and prognosis
Small cell	Small round cells producing lace-like osteoid; may be CD99 positive	**Main differential is Ewing sarcoma** (no EWSR1 translocation); treated as conventional; slightly worse prognosis
Low-grade central	Metaphysis of femur or tibia, adults 20-40; slowly growing; **mimics fibrous dysplasia**; **MDM2 and CDK4 amplification**	**Wide excision alone**; no chemotherapy; excellent prognosis unless it dedifferentiates
Parosteal	Commonest surface OS (about 5% of all OS); **posterior distal femoral metaphysis** (popliteal surface); women 20-40 years; dense, lobulated mass **"wrapped around" the bone**; **radiolucent cleavage line (string sign)** between tumour and cortex; **MDM2/CDK4 amplification**; low grade	**Wide excision alone**; 5-year survival about 90%; marginal excision recurs; **dedifferentiated parosteal OS** (about 15-25%) is high-grade and treated as conventional
Periosteal	Surface of the **diaphysis of tibia or femur**, adolescents; **predominantly chondroblastic**, intermediate grade; saucerised cortex with **perpendicular "hair-on-end" spicules** and Codman triangle; medulla spared	Wide excision; **role of chemotherapy debated** (often given); prognosis better than conventional, worse than parosteal
High-grade surface	Rare, surface lesion but histologically like conventional	As conventional (chemotherapy + surgery)
Secondary (Paget, irradiation, infarct)	Older patients	Chemotherapy often poorly tolerated; poor prognosis
Extraskeletal osteosarcoma	Soft tissue of thigh in adults over 50	Treated like soft tissue sarcoma; poor prognosis

- **Parosteal versus myositis ossificans (most important differential):**

- **Myositis ossificans: zonal pattern** with **mature bone at the periphery and immature centre**; history of trauma; matures over 6-8 weeks; separated from cortex.
- **Parosteal osteosarcoma: densest centrally at the base**, less mature at the periphery (reverse zonal pattern); attached to cortex by a broad base.
- Also differentiate from osteochondroma (corticomedullary continuity, cartilage cap) and **BPOP** (bizarre parosteal osteochondromatous proliferation, hands and feet).

Exam pearl: Parosteal OS = posterior distal femur, woman 20-40, "string sign", MDM2 amplification, low grade, wide excision only, about 90% survival. Periosteal OS = tibial diaphysis, chondroblastic, hair-on-end. Telangiectatic OS = fluid-fluid levels like ABC but with thick nodular septa - never curette an "ABC" without histology.

7. Summary of radiological differentials for a bone-forming lesion

Lesion	Discriminating feature
Stress fracture	Linear periosteal/endosteal reaction, no soft tissue mass, history of activity
Exuberant callus	History of fracture, matures over weeks
Osteomyelitis	Sequestrum, cloaca, systemic features; can mimic Ewing or OS
Ewing sarcoma	Diaphyseal, onion-skin periosteal reaction, large soft tissue mass, no osteoid matrix
Osteoblastic metastasis	Older patient, multiple lesions (prostate, breast)
Paget disease	Coarsened trabeculae, cortical thickening, bone enlargement, starts at an end of the bone and advances (blade of grass / flame-shaped front in long bones)

Cartilage tumours

1. Overview

- **Chondrogenic tumours** form hyaline cartilage matrix. Radiographic hallmark of cartilage matrix: **"rings and arcs" / "popcorn" / punctate (stippled) calcification**.
- **WHO 2020 classification (bone):**
  - **Benign:** osteochondroma, enchondroma, periosteal chondroma, chondroblastoma, chondromyxoid fibroma, subungual exostosis, bizarre parosteal osteochondromatous proliferation (BPOP), synovial chondromatosis, osteochondromyxoma.
  - **Intermediate (locally aggressive): atypical cartilaginous tumour (ACT)** of the appendicular skeleton (formerly grade 1 chondrosarcoma).
  - **Malignant:** chondrosarcoma grades 1-3 (grade 1 term kept for **axial skeleton**: pelvis, scapula, skull base), periosteal chondrosarcoma, **clear cell**, **mesenchymal** and **dedifferentiated** chondrosarcoma.
- **Location in bone (classic exam table):**

Location	Cartilage lesion
Epiphysis, skeletally immature	**Chondroblastoma**
Epiphysis, adult	**Clear cell chondrosarcoma** (and GCT, which is not cartilaginous)
Metaphysis, eccentric	**Chondromyxoid fibroma**, osteochondroma
Medullary metaphysis/diaphysis	Enchondroma, central chondrosarcoma
Surface	Osteochondroma, periosteal chondroma, periosteal chondrosarcoma
Short tubular bones of the hand	Enchondroma (commonest primary bone tumour of the hand)

- **Molecular high-yield: IDH1/IDH2 mutations** in enchondroma, Ollier, Maffucci and about 50% of central (conventional and dedifferentiated) chondrosarcomas; **EXT1/EXT2** in osteochondroma and HME; **H3F3B K36M**

(H3.3) in chondroblastoma; **GRM1** rearrangement in chondromyxoid fibroma; **HEY1-NCOA2** fusion in mesenchymal chondrosarcoma.

2. Osteochondroma (osteocartilaginous exostosis)

Pathology and epidemiology

- **Commonest benign bone tumour** (about one third of benign bone tumours; many unrecognised). Age 10-30, slight male predominance.
- A **developmental lesion**: physeal cartilage herniates through the periosteal bone collar (perichondral ring of LaCroix) and grows at right angles to the bone.
- **Defining feature: cortical and medullary continuity with the host bone**, capped by hyaline cartilage. It grows by enchondral ossification under the cap and **stops growing at skeletal maturity**.
- **Sites:** metaphysis of long bones, especially **distal femur, proximal tibia (around the knee) and proximal humerus**; points **away from the adjacent joint** (grows along tendon pull). Sessile or pedunculated.
- Radiation can induce osteochondromas (e.g. after childhood irradiation).

Clinical features and complications

- Painless hard mass; pain from **overlying bursa (bursitis)**, tendon snapping, **fracture of the stalk**.
- **Neurovascular compression: common peroneal nerve** at the proximal fibula; **popliteal artery pseudoaneurysm** or thrombosis from distal femoral or proximal tibial lesions.
- Growth disturbance and deformity (especially in multiple exostoses), spinal cord compression (rare spinal lesions).
- **Malignant transformation to secondary peripheral chondrosarcoma: about 1% for solitary lesions**; **about 1-5% (lifetime) in HME** (older figures up to 10-20% are overestimates; debated). Pelvis, scapula and proximal femur lesions are the highest risk.
- **Warning signs of malignancy: growth or new pain after skeletal maturity**, **cartilage cap thicker than 2 cm in an adult** (some use 1.5 cm) on MRI, irregular cap with lucent areas or soft tissue mass with scattered calcification, cortical destruction of the stalk.
  - In growing children, caps up to 3 cm can be normal.

Treatment

- **Observation** for asymptomatic lesions.
- **Excision** for pain, neurovascular compression, cosmesis, deformity or suspicion of malignancy: **remove the whole cartilage cap with its perichondrium** (and stalk base) to prevent recurrence. Ideally after skeletal maturity to reduce recurrence and avoid physeal damage.
- **Peroneal nerve** must be identified and protected when excising proximal fibular lesions.
- Suspected malignant change: biopsy-guided **wide en bloc resection** (do not shell out).

Hereditary multiple exostoses (HME, multiple osteochondromas)

- **Autosomal dominant**, high penetrance (about 96%); **EXT1 (chromosome 8q24)** gives a more severe phenotype than **EXT2 (11p)**. EXT genes encode glycosyltransferases for **heparan sulphate** synthesis.
- **Short stature**, limb length discrepancy, **genu valgum** (proximal tibia), **ankle valgus** (distal tibia and fibular shortening), **coxa valga** and hip subluxation.
- **Forearm deformity (in 40-60%)**: the **ulna is short** (lesions at the distal ulna), radius bowed with ulnar inclination of the distal radius, ulnar translation of the carpus, and **radial head dislocation**.
- **Masada classification of the forearm:**

Type	Description	Management idea
I	Osteochondroma of distal ulna; ulna short; radius bowed; radial head **not** dislocated	Excision + ulnar lengthening (gradual distraction) ± radial corrective osteotomy
IIa	Radial head dislocated, with an osteochondroma of the proximal radius	Excision; radial head excision at maturity if painful
IIb	Radial head dislocated secondary to a short ulna (distal ulnar lesion), no proximal radial lesion	Ulnar lengthening ± open reduction of radial head (difficult)
III	Osteochondroma of distal radius with relative shortening of the radius	Excision

- **Treatment principles in HME:** excise symptomatic lesions, **guided growth (hemiepiphysiodesis)** for genu valgum and ankle valgus, lengthening for forearm and limb discrepancy, lifelong awareness of malignant change (new pain or growth in adulthood).
- **Related conditions:**
  - **Trevor disease (dysplasia epiphysealis hemimelica):** intra-articular **epiphyseal osteochondroma**, usually on the **medial side** of one lower limb (talus, distal tibia, distal femur); boys about 3:1. Excise symptomatic lesions; preserve articular surface.
  - **Metachondromatosis:** AD, **PTPN11**; both exostoses (which **point towards the joint**, mainly hands and feet) and enchondromas; lesions may regress.
  - **Subungual exostosis (Dupuytren):** distal phalanx of the great toe, trauma-related, **no medullary continuity**; excise with cap.
  - **BPOP (Nora lesion):** surface lesion of hands and feet, no corticomedullary continuity, high recurrence.

> Exam pearl: Osteochondroma = corticomedullary continuity + cartilage cap, points away from the joint, stops growing at maturity. Cap over 2 cm in an adult or growth/pain after maturity = suspect chondrosarcoma. HME = AD, EXT1 (8q24, worse) / EXT2 (11p), heparan sulphate, short ulna with radial head dislocation (Masada).

3. Enchondroma and enchondromatoses

Solitary enchondroma

- Benign intramedullary hyaline cartilage, probably from physeal rests. Age 20-50. Often an incidental finding.
- **Commonest primary bone tumour of the hand** (about half occur in the hand): **proximal phalanx** most common, then metacarpals and middle phalanges. Also proximal humerus, distal femur and proximal tibia (metaphysis to diaphysis).
- **Hand:** presents with **pathological fracture** after minor trauma; expansile lytic lesion with thinned cortex and stippled calcification.

- **Management:** allow the fracture to heal (about 4-6 weeks), then **curettage ± bone graft** or bone substitute (some treat in one stage). Recurrence is uncommon. Malignant change in the hand is very rare.

- **Long bones:** central lesion with **rings-and-arcs calcification**, little or no endosteal scalloping, no cortical breach, no soft tissue mass, usually painless.
  - Painless lesions without aggressive features: **observation with serial imaging** (radiographs or MRI).

Enchondroma versus atypical cartilaginous tumour / low-grade chondrosarcoma in long bones

Feature	Favours enchondroma	Favours ACT / chondrosarcoma
Pain	Absent (pain usually from another cause, e.g. knee OA)	**Pain attributable to the lesion**, especially at night
Endosteal scalloping	Absent or shallow	**Deep (more than two thirds of cortical thickness)** and long (over two thirds of lesion length)
Cortex	Normal	**Cortical thickening, expansion, breach**
Periosteal reaction / soft tissue mass	Absent	Present
Size	Usually under 5 cm	Larger (often over 5 cm)
Bone scan uptake	Mild	Greater than the anterior iliac crest
Site	Hand, long bones	**Axial skeleton (pelvis) almost always malignant**
Histology	Hypocellular lobules **encasing** host trabeculae	**Permeation and entrapment of host trabeculae**, myxoid change, increased cellularity

- Histology alone is unreliable for distinguishing enchondroma from low-grade chondrosarcoma (high interobserver variability); **clinical, radiological**

and pathological correlation is required.

- **Bone infarct** (differential): **serpiginous peripheral sclerotic rim** ("smoke rising up a chimney"), no scalloping, no central rings and arcs.

Ollier disease and Maffucci syndrome

Condition	Features	Malignancy risk
Ollier disease (enchondromatosis)	**Multiple enchondromas**, often **unilateral predominance**, sporadic (not inherited), **somatic mosaic IDH1/IDH2** mutations; limb shortening, angular deformity, short ulna with forearm bowing, hand deformity	**Secondary central chondrosarcoma in about 25-40%** (figures vary; debated)
Maffucci syndrome	Multiple enchondromas + **soft tissue (spindle cell) haemangiomas** (blue nodules with **phleboliths** on radiographs); IDH1/IDH2	**Higher (up to about 50%)**, plus **non-skeletal malignancies** (gliomas, ovarian juvenile granulosa cell tumour, pancreatic and hepatic tumours)

- Lifelong surveillance; any new pain or growth needs MRI and evaluation.
- Limb deformity: corrective osteotomies, gradual lengthening (Ilizarov / hexapod); lesions within distraction regenerate ossify.

Periosteal (juxtacortical) chondroma

- Surface cartilage lesion under the periosteum, young adults, **proximal humerus** commonest, hand.
- **Saucerisation of the cortex with a sclerotic buttress**, no medullary involvement; usually **under 3 cm**.
- Differential: **periosteal chondrosarcoma** (larger, usually over 3-5 cm, older patients, invasion of cortex/medulla).
- Treatment: **marginal en bloc excision** with the underlying cortex; low recurrence.

4. Chondroblastoma

- About 1% of bone tumours. **Age 10-25 (skeletally immature)**; male predominance about 2:1.
- **Epiphysis or apophysis** (equivalent): **proximal humerus (Codman tumour)**, **distal femur**, **proximal tibia**, **greater trochanter**; also **talus and calcaneus** in older patients.
- **Symptoms:** joint pain, effusion, stiffness (mimics intra-articular pathology).
- **Radiograph:** small, round, well-defined **lytic epiphyseal lesion with a thin sclerotic rim**, central calcification in about a third to half; may cross the physis into the metaphysis.
- **MRI: extensive surrounding marrow and soft tissue oedema** and periosteal reaction (out of proportion to the lesion); **secondary ABC** change in about 15-30% (fluid-fluid levels).
- **Histology:** polygonal **chondroblasts with grooved "coffee-bean" nuclei**, scattered osteoclast-like giant cells, pink chondroid matrix, **"chicken-wire" pericellular calcification**. Immunohistochemistry: S100, SOX9, DOG1 and **H3 K36M mutant-specific antibody** (H3F3B).
- **Treatment: extended curettage** (high-speed burr ± phenol, cryotherapy or argon adjuvant) and grafting or cementing, approached to protect the **physis and articular cartilage** (e.g. through the metaphysis or via a cortical window). **Local recurrence about 10-20%** (higher in pelvis, hip, talus, and with secondary ABC). RFA for small lesions in difficult sites.
- **Benign lung metastases** in about 1-2% (usually after local recurrence); treated by excision, good prognosis.
- **Differentials (epiphyseal lucent lesions):** GCT (skeletally mature, no sclerotic rim, extends to subchondral bone), clear cell chondrosarcoma (adult), infection (Brodie abscess can be epiphyseal in children), LCH, intraosseous ganglion, subchondral cyst/geode.

5. Chondromyxoid fibroma (CMF)

- **Rarest benign cartilage tumour** (under 1% of bone tumours). Age 10-30.
- **Metaphysis of the proximal tibia** is the classic site (about one third around the knee), also distal femur, pelvis, feet.

- **Radiograph: eccentric, elongated, lobulated lytic metaphyseal lesion** with a **sclerotic scalloped ("bite-like") margin**, cortical expansion, **calcification uncommon** (unlike other cartilage lesions).
- **Histology:** lobules of **stellate and spindle cells in myxoid and chondroid stroma**, hypocellular centres with hypercellular periphery; **GRM1** rearrangement.
- **Differentials:** NOF (also eccentric metaphyseal, but cortical, multiloculated, children), ABC, fibrous dysplasia, chondrosarcoma (histological mimic).
- **Treatment: extended curettage with adjuvant and bone grafting**; recurrence about 15-25%; en bloc resection for expendable bones.

6. Chondrosarcoma

Epidemiology and classification

- **Second commonest primary malignant bone tumour** (about 20-30%). **Adults 40-70 years**, slight male predominance. Rare in children (consider mesenchymal subtype).
- **Sites: pelvis (ilium) commonest**, then **proximal femur**, **proximal humerus**, ribs and scapula. Flat bones and proximal limb bones (unlike enchondroma, which favours the hand).
- **Classification by origin:**
  - **Primary central (conventional)**: about 85-90%; arises de novo in the medulla.
  - **Secondary central**: from enchondroma (Ollier, Maffucci).
  - **Secondary peripheral**: from the cap of an osteochondroma (solitary or HME); usually low grade; treated by wide excision.
  - **Periosteal chondrosarcoma**: surface.
- **Subtypes:** dedifferentiated (about 10%), clear cell (about 2%), mesenchymal (about 2-3%).

Clinical and radiology

- **Pain (often deep, dull, nocturnal) is the key symptom**; mass in pelvic lesions often large at presentation; pathological fracture in high grade.
- **Radiograph and CT:** lytic expansile lesion with **rings-and-arcs calcification**, **deep endosteal scalloping**, **cortical thickening**, cortical destruction,

periosteal reaction, soft tissue mass. A **purely lytic area within a calcified cartilage lesion suggests dedifferentiation** (target the biopsy there).

- **MRI:** lobulated, **very high T2 signal** (hyaline cartilage), peripheral and septal ("ring and arc") enhancement; defines extent and soft tissue mass.
- **Staging:** CT chest, bone scan or PET-CT. Biopsy from the most aggressive area (CT/MRI guided), at the treating centre.

Grading and prognosis

Grade	Histology	Metastasis	Approximate 10-year survival
ACT / grade 1	Low cellularity, mild atypia, abundant hyaline matrix	Rare	About 90%
Grade 2	More cellular, myxoid change, nuclear atypia, mitoses rare	About 10-30%	About 60-70%
Grade 3	Highly cellular, pleomorphic, mitoses, myxoid	Up to 70%	About 30-50%
Dedifferentiated	Low-grade cartilage **abruptly juxtaposed with high-grade non-cartilaginous sarcoma** (osteosarcoma, UPS, fibrosarcoma)	High	**5-year survival about 10-25%**

- **Grade is the most important prognostic factor**; others: margin status (local recurrence), axial site, size.
- Lungs are the commonest metastatic site.

Management

- **Conventional chondrosarcoma is resistant to chemotherapy and radiotherapy** (low mitotic rate, poor vascularity, abundant matrix), so **surgery is the mainstay**.
- **ACT of long bones (intracompartmental, no soft tissue extension): extended intralesional curettage with local adjuvant** (high-speed burr, phenol, liquid nitrogen cryotherapy, argon, or ethanol) and cement or graft; local recurrence low (about 5%), with better function than resection. Small

asymptomatic lesions are increasingly managed by **active surveillance** (debated).

- **Grade 1 axial (pelvis, scapula, skull base), grade 2, grade 3, and secondary peripheral chondrosarcoma: wide en bloc resection**. Intralesional surgery leads to recurrence, and recurrent tumours can be of higher grade.
  - **Pelvic chondrosarcoma (Enneking-Dunham pelvic resection zones): I = ilium, II = periacetabular, III = pubis and ischium (obturator ring), IV = sacral extension**. Periacetabular (zone II) resections require reconstruction (custom or modular pelvic prosthesis, ice-cream-cone implants, allograft, hip transposition) or are left as a flail hip; **hindquarter amputation** if wide margins cannot otherwise be achieved.
- **Radiotherapy:** for incomplete resection, inoperable sites (skull base, spine) using high-dose **proton or carbon-ion therapy**; palliation.
- **Chemotherapy:** only for **mesenchymal** (Ewing-type regimens) and **dedifferentiated** chondrosarcoma (osteosarcoma-type regimens; benefit debated), and metastatic disease in trials. **IDH1 inhibitors (ivosidenib)** are under study for IDH-mutant tumours.
- **Follow-up:** long term (10 years or more) because of late local recurrence and lung metastases.

Other subtypes

Subtype	Key features	Treatment
Clear cell chondrosarcoma	Low grade; **epiphysis of proximal femur (about two thirds) or proximal humerus**; adults 25-50, males; well-defined lytic epiphyseal lesion; **mimics chondroblastoma** but in a skeletally mature patient; clear cells with osteoclast-like giant cells	**Wide en bloc resection** (curettage leads to high recurrence); late metastases possible, long follow-up

Subtype	Key features	Treatment
Mesenchymal chondrosarcoma	Young (10-30); jaw, ribs, spine, pelvis, and **extraskeletal** (meninges); **biphasic: undifferentiated small round blue cells with islands of hyaline cartilage**, haemangiopericytoma-like vessels; **HEY1-NCOA2** fusion; CD99 positive (differential: Ewing)	Wide resection + **chemotherapy** (Ewing-type) ± radiotherapy; poor prognosis, late metastases
Dedifferentiated chondrosarcoma	Older (50-70); pelvis, femur, humerus; **bimorphic** pattern; lytic aggressive component next to calcified cartilage; pathological fracture common	Wide resection or amputation ± chemotherapy; very poor prognosis
Periosteal chondrosarcoma	Surface of femur/humerus; usually **over 5 cm**; invades cortex	Wide excision; good prognosis (low grade)

Exam pearl: Chondrosarcoma: adult, pelvis commonest, painful lesion with rings-and-arcs calcification and deep endosteal scalloping. Chemo- and radio-resistant; grade is the key prognostic factor. Appendicular ACT = extended curettage + adjuvant; axial grade 1 and all grade 2/3 = wide resection. Clear cell CS = adult epiphysis of proximal femur (chondroblastoma look-alike). Mesenchymal CS = small round cells + cartilage, HEY1-NCOA2, chemotherapy-sensitive. Dedifferentiated = bimorphic, worst prognosis.

7. Synovial chondromatosis

- Classified with cartilage tumours (WHO 2020) as a benign, locally aggressive neoplasm of the synovium (**FN1-ACVR2A / FN1 rearrangements**).
- Monoarticular, adults 30-50, **knee commonest**, then hip, elbow, shoulder.
- Multiple cartilaginous nodules in the synovium that detach as loose bodies and may calcify (**multiple uniform-sized "rice-grain" ossified loose bodies**); erosion of bone on both sides of the joint.
- **Milgram stages: I** active intrasynovial disease, no loose bodies; **II** transitional, synovial nodules and loose bodies; **III** multiple loose bodies, quiescent synovium.

- **Treatment: loose body removal + synovectomy** (open or arthroscopic); recurrence about 10-20%. Malignant transformation to chondrosarcoma is rare (under about 5%), suspected with rapid recurrence or marrow invasion.
- Differentials: PVNS (haemosiderin, blooming on gradient-echo MRI, no calcification), secondary osteochondromatosis (loose bodies of varying size from OA).

8. Quick comparison table of cartilage tumours

Lesion	Age	Site	Discriminating clue
Osteochondroma	10-30	Metaphysis around knee	Corticomedullary continuity, points away from joint
Enchondroma	20-50	Hand phalanges, long bone medulla	Pathological fracture of proximal phalanx; rings-and-arcs
Chondroblastoma	10-25	**Epiphysis**, proximal humerus	Chicken-wire calcification, coffee-bean nuclei, oedema on MRI
Chondromyxoid fibroma	10-30	Proximal tibial metaphysis, eccentric	Scalloped sclerotic rim, little calcification
Chondrosarcoma	40-70	Pelvis, proximal femur/humerus	Pain, deep endosteal scalloping, cortical thickening
Clear cell CS	25-50	Epiphysis proximal femur	Adult "chondroblastoma"
Mesenchymal CS	10-30	Jaw, ribs, extraskeletal	Small round cells + cartilage

Giant cell and cystic lesions

1. Approach to a lytic (lucent) bone lesion

- **Four questions:** age of patient, location in the bone (epiphysis, metaphysis, diaphysis; central, eccentric, cortical), margin (zone of transition), matrix and periosteal reaction. Number of lesions.
- **Lodwick classification of margins (rate of growth):**

Grade	Pattern	Meaning
IA	Geographic, **sclerotic rim**	Slow growing, benign (NOF, UBC, fibrous dysplasia)
IB	Geographic, well-defined, **no sclerotic rim**	Faster (GCT, myeloma)
IC	Geographic, **ill-defined margin**	Aggressive
II	**Moth-eaten**	Aggressive (infection, lymphoma, metastasis)
III	**Permeative** (wide zone of transition)	Very aggressive: Ewing, lymphoma, osteomyelitis

- **Epiphyseal lucent lesions:** chondroblastoma (immature), **GCT** (mature), clear cell chondrosarcoma, infection, LCH, intraosseous ganglion, subchondral cyst (geode).
- **Multiple lesions in an adult:** metastases, myeloma, brown tumours, polyostotic fibrous dysplasia, enchondromatosis.
- **Fluid-fluid levels on MRI (not specific): ABC (primary or secondary), telangiectatic osteosarcoma**, GCT, chondroblastoma, UBC after fracture.
- **Giant-cell-rich lesions (histological differential):** GCT, **brown tumour of hyperparathyroidism**, ABC, chondroblastoma, NOF, giant cell reparative granuloma (jaw, hands and feet), giant-cell-rich osteosarcoma, pigmented villonodular synovitis extending into bone.
- **Rule:** a "benign-looking" cystic lesion with solid enhancing areas, thick nodular septa, cortical destruction or soft tissue mass needs a **biopsy before curettage** (telangiectatic OS can mimic ABC).

2. Giant cell tumour of bone (GCTB)

Epidemiology and pathology

- About **5% of primary bone tumours** and about 20% of benign bone tumours; **more common in Asian (including Indian and Chinese) populations**.
- **Age 20-40, after physeal closure** (about 80%); rare under 15 (if so, may be metaphyseal). **Slight female predominance**.
- **Sites: distal femur > proximal tibia > distal radius** (third commonest, and most aggressive with highest recurrence) > **sacrum** (commonest axial site; the vertebral body above the sacrum is uncommon) > proximal humerus. About 50-60% around the knee. Multicentric GCT is rare (under 1%; consider hyperparathyroidism and hand involvement).
- **Pathology:**
  - **Neoplastic cells = mononuclear stromal cells** (osteoblast-lineage), which express **RANKL**; they carry the **H3F3A (H3.3) G34W mutation in over 90%** (immunostain H3.3 G34W is diagnostic).
  - **Osteoclast-like giant cells** (expressing **RANK**) are reactive; they are recruited by RANKL and resorb bone. Their nuclei resemble the stromal cell nuclei.
  - **This RANKL-RANK pathway is the basis of denosumab therapy.**
- Classified by WHO as **intermediate, locally aggressive, rarely metastasising**.

Clinical

- Pain, swelling near a joint, effusion, reduced movement; **pathological fracture in about 10-30%**.

Imaging

- **Eccentric, lytic, epiphyseo-metaphyseal lesion extending up to the subchondral bone** (within about 1 cm of the articular surface).
- **Geographic, non-sclerotic margin (Lodwick IB)**, no matrix mineralisation, cortical expansion and thinning, "soap-bubble" appearance from ridges, cortical breach and soft tissue mass in aggressive cases. Periosteal reaction unusual.

- **MRI:** low-to-intermediate T2 signal in solid areas (haemosiderin and collagen), fluid-fluid levels from secondary ABC change (about 15%); defines soft tissue extension and joint involvement.
- **CT chest at diagnosis** (lung metastases).
- **Bloods: serum calcium, phosphate and PTH** to exclude **brown tumour of hyperparathyroidism**, especially with multiple lesions, unusual sites or giant cells in the jaw/hands.

Campanacci radiographic grading (corresponds to Enneking benign stages 1-3)

Grade	Radiographic features	Implication
I (latent)	Well-marginated, **thin sclerotic rim**, cortex intact	Rare; extended curettage
II (active)	Well-defined margin **without sclerotic rim**, cortex thinned and expanded but intact	Commonest (about 70-80%); **extended curettage + adjuvant + cement**
III (aggressive)	**Indistinct margins, cortical breach, soft tissue mass**	Higher recurrence; curettage still possible if joint salvageable, otherwise **en bloc resection** and reconstruction

Management

Extended (intralesional) curettage: treatment of choice for most

1. **Large cortical window** (as big as the lesion) so every recess can be seen directly.
2. Thorough **curettage**, then **high-speed burr** of all walls.
3. **Local adjuvant** to extend the margin: **phenol**, **liquid nitrogen cryotherapy**, **argon beam coagulation**, hydrogen peroxide or absolute alcohol; pulsed lavage.
4. **Filling the cavity with PMMA bone cement**: exothermic polymerisation kills residual cells (about 2-3 mm), gives immediate structural support and allows early weight bearing, and makes **recurrence easy to detect** (expanding lucency at the cement-bone interface). Subchondral bone graft

beneath the cement to protect articular cartilage is optional (debated). Internal fixation (plate) if the defect weakens the bone.

5. **Recurrence rates:** curettage alone historically **about 40-60%**; extended curettage with burr and adjuvant **about 10-20%**. Most recurrences appear **within 2-3 years**; distal radius, Campanacci III and soft tissue extension recur more often.
6. **Pathological fracture** is not an automatic indication for resection; if the fracture is not grossly displaced into the joint, curettage can be done (after or with fixation).

En bloc (wide) resection

- **Indications:** extensive destruction with an unsalvageable joint, large soft tissue component (Campanacci III), grossly displaced intra-articular pathological fracture, multiply recurrent tumour, and **expendable bones** (proximal fibula, distal ulna, rib, iliac wing, coccyx).
- Lower recurrence (about 0-10%) but worse function.
- **Distal radius Campanacci III:** en bloc resection + reconstruction with **non-vascularised or vascularised proximal fibula autograft (osteoarticular, fibula head to the carpus)**, osteoarticular allograft, or **ulnar translocation with wrist arthrodesis** (radiocarpal fusion).
- Around the knee: megaprosthesis or osteoarticular allograft.

Spine and sacrum

- Sacral and spinal GCT are difficult: en bloc resection risks bowel, bladder and sexual dysfunction (sacral nerve roots).
- Options: **intralesional curettage + adjuvant with pre-operative embolisation**, **serial arterial embolisation**, **denosumab**, and **radiotherapy** (now reserved for unresectable disease because of the risk of **radiation-induced malignant transformation**, historically up to about 10%, lower with modern megavoltage techniques).

Medical therapy

- **Denosumab** (fully human monoclonal antibody against **RANKL**):
  - Dose: **120 mg subcutaneously every 4 weeks, with loading doses on days 8 and 15 of the first month**, with calcium and vitamin D supplements.

- **FDA approved (2013)** for adults and skeletally mature adolescents with **unresectable GCT or where surgery would cause severe morbidity**, based on open-label phase 2 trials (Thomas et al., Lancet Oncology 2010; Chawla et al., 2013 and 2019) showing tumour response in most patients (loss of giant cells, new bone formation).
- **Uses:** unresectable or sacral/spinal tumours, metastatic disease, and **neoadjuvant downstaging** to allow joint-preserving surgery (debated).
- **Problems:**
- **Higher local recurrence after curettage when used pre-operatively** (thick new bone and sclerotic rim hide residual tumour; stromal cells are not killed) - so for lesions already suitable for curettage, **go straight to surgery**.
- **Recurrence or regrowth after stopping**, so long-term treatment may be needed (optimal duration unknown; dosing intervals are often extended).
- **Osteonecrosis of the jaw** (about 1-5%, increasing with duration), **hypocalcaemia**, **atypical femoral fractures**, rebound hypercalcaemia in young patients after stopping, and **reports of malignant transformation** (sarcoma) during treatment.

- **Bisphosphonates (zoledronic acid)** also inhibit osteoclast giant cells; used as a cheaper adjuvant to reduce recurrence and for unresectable disease (evidence weaker).
- **Interferon alfa** and radiotherapy for refractory cases.

Metastases and malignancy

- **Lung metastases ("benign pulmonary implants") in about 2-5%**, usually 2-4 years after diagnosis and often after local recurrence (especially distal radius). Histologically identical to the primary; usually indolent. Treat by **metastasectomy**, observation or denosumab; good prognosis.
- **Malignancy in GCT (about 1-4%):**
  - **Primary malignant GCT:** high-grade sarcoma juxtaposed with conventional GCT at presentation.
  - **Secondary malignant GCT:** sarcoma at the site of previous GCT, usually **after radiotherapy** (or rarely after surgery or denosumab). Poor prognosis; treated like osteosarcoma.

- **Follow-up:** radiographs of the site and **chest imaging every 3-6 months for 2-3 years**, then annually to about 5-10 years.

> Exam pearl: GCT = skeletally mature, eccentric epiphyseo-metaphyseal lytic lesion reaching subchondral bone, non-sclerotic margin. Neoplastic cell = RANKL-expressing stromal cell (H3F3A G34W); giant cells are reactive. Distal femur > proximal tibia > distal radius (most aggressive); sacrum commonest axial site. Extended curettage + burr + adjuvant + PMMA (recurrence about 10-20%). Denosumab for unresectable disease - not routinely before curettage (more recurrence). Lung implants 2-5%. Always check calcium/PTH to exclude brown tumour.

3. Unicameral (simple) bone cyst (UBC)

Features

- Fluid-filled cavity lined by a thin fibrous membrane; **not a true neoplasm** (although **NFATC2 fusions** have recently been described). Cause thought to be **venous obstruction** with fluid rich in prostaglandins, interleukin-1 and enzymes that resorb bone.
- **Age 5-15 years, male to female about 2:1**.
- **Sites: proximal humerus (about 50-60%)**, **proximal femur (about 25-30%)**; in adults, **calcaneus** (neutral triangle under the angle of Gissane) and ilium.
- **Pathological fracture is the commonest presentation (about two thirds to three quarters)**; otherwise incidental.

Radiology

- **Central (not eccentric) metaphyseal lytic lesion**, **symmetrically expanded but not wider than the physis**, thin sclerotic margin, thinned cortex, no periosteal reaction without fracture, no matrix.
- **"Fallen fragment" sign** (a cortical fragment sinks to the dependent part of the fluid-filled cyst) is **pathognomonic** of UBC (versus solid lesions). The "rising bubble" sign is the gas equivalent.
- **Active vs latent:**

Phase	Features	Prognosis
Active	**Within about 1 cm of (abutting) the physis**, usually **under 10-12 years**	High recurrence after treatment; risk of physeal growth arrest
Latent (inactive)	**Migrated away from the physis** into the diaphysis by normal growth (over 1 cm), older child	Lower recurrence; tends to heal

- **Risk of fracture:** cyst index (cyst area / diaphyseal diameter squared) and thin cortex; fracture is likely when the cyst occupies most of the bone width.
- **Calcaneal UBC versus intraosseous lipoma:** lipoma has fat density and often a **central calcification ("cockade" / target)**; UBC has fluid content.
- **Differentials:** ABC (eccentric, expands wider than the physis, fluid-fluid levels, multiple septa), fibrous dysplasia (ground glass), LCH, enchondroma, NOF.

Management

- **Fracture through a proximal humeral cyst:** treat the fracture non-operatively (sling); the fracture heals in about 6 weeks, but **the cyst heals after fracture in only about 10-15%**. Then decide on treatment of the cyst.
- **Observation** in older children with latent lesions and low fracture risk.
- **Minimally invasive options:**
  - **Aspiration + intralesional methylprednisolone acetate injection** (Scaglietti 1979); often **multiple injections** needed. Confirm with straw-coloured fluid and contrast cystogram (contrast stays in a single cavity).
  - Injection of **autologous bone marrow** or **demineralised bone matrix**, or calcium sulphate pellets.
  - **Decompression** (trephination / curettage with disruption of the cyst wall and canal communication) and **flexible intramedullary nails** (elastic stable intramedullary nails act as a continuous drain and give stability; allow early return to activity), or cannulated screws.
- **Evidence: Wright et al. (JBJS Am 2008), multicentre RCT: methylprednisolone injection gave a higher healing rate (about 42%) than bone marrow injection (about 23%)** at 2 years. Overall, no technique

gives complete healing in every case; recurrence after any method is common (about 20-40%).

- **Open curettage and bone grafting** with or without adjuvant: for failed injections or large lesions; risk of **physeal injury** in active cysts.
- **Proximal femoral UBC:** high risk of fracture with **varus malunion, AVN and growth arrest** - so treat **surgically**: curettage + graft and **internal fixation** (flexible nails, pediatric hip screw or plate, avoiding the physis), especially for large cysts or after fracture.
- **Assessment of healing** (Neer/Cole type classifications): healed, healed with residual radiolucency, persistent, recurrent.

> Exam pearl: UBC = 5-15 years, proximal humerus, central metaphyseal, not wider than the physis, fallen fragment sign pathognomonic. Active = abutting the physis (under 1 cm), recurs more. Fracture heals but the cyst rarely heals. Steroid injection beat bone marrow injection in the Wright 2008 RCT. Proximal femur cysts need surgical fixation.

4. Aneurysmal bone cyst (ABC)

Features

- **Blood-filled, multiloculated expansile lesion**; WHO intermediate, locally aggressive.
- **Age under 20 in about 80%** (peak 10-20), roughly equal sex distribution.
- **Sites:** metaphysis of long bones (**femur, tibia, humerus**), **spine (posterior elements, can extend into the body; about 15-20%)**, pelvis, sacrum.
- **Primary ABC (about 70%)**: true neoplasm with **USP6 gene rearrangement** (17p13; commonest partner **CDH11**). Also seen in **nodular fasciitis and myositis ossificans** (USP6 family).
- **Secondary ABC (about 30%)**: ABC-like change within another lesion, without USP6 rearrangement - **GCT** (commonest), **chondroblastoma**, **osteoblastoma**, fibrous dysplasia, NOF, UBC, and **telangiectatic osteosarcoma**. Treatment is that of the underlying lesion.
- **Solid variant (giant cell reparative granuloma-like)** is part of the same spectrum.

Clinical and imaging

- Pain and swelling of short duration, pathological fracture; spine: back pain, scoliosis, **radiculopathy or cord compression**.
- **Radiograph: eccentric, expansile, "blow-out" lytic lesion** with a thin **eggshell** cortical rim, internal trabeculation ("soap bubble"), may expand **wider than the physis**; Codman triangle may be seen at the margins.
- **MRI: multiple fluid-fluid levels** (blood products of differing ages), **thin, smooth enhancing septa**.
- **Telangiectatic osteosarcoma must be excluded:** thick nodular enhancing septa, solid component, cortical destruction, soft tissue mass and aggressive margins favour TOS. **Biopsy of the septa and solid areas** before definitive treatment.
- **Histology:** blood-filled cavernous spaces separated by **fibrous septa containing fibroblasts, osteoclast-like giant cells and "blue bone"** (reactive woven bone); no endothelial lining.

Management

Option	Notes
Extended curettage + adjuvant (burr, phenol, cryotherapy, argon) + bone graft or cement	Standard; **recurrence about 10-20%**, higher in young children and lesions near the physis; most recurrences within 2 years
Percutaneous sclerotherapy (**polidocanol**, doxycycline, alcohol)	Minimally invasive; multiple injections; an RCT (**Varshney et al. 2010**) found polidocanol gave healing comparable to curettage with fewer complications; increasingly used first line, especially in difficult sites
Selective arterial embolisation	Pelvis, sacrum and spine; as primary treatment or to reduce intra-operative bleeding before surgery
Denosumab	Unresectable spinal or sacral lesions; regrowth after stopping and **rebound hypercalcaemia in children** are concerns
En bloc resection	Expendable bones (fibula, rib) or recurrent aggressive lesions

Option	Notes
Spine	Pre-operative embolisation, curettage or excision, decompression and **instrumented stabilisation** for instability or deficit
Radiotherapy	**Avoided** (risk of radiation-induced sarcoma, growth disturbance); last resort

Exam pearl: Primary ABC = USP6 (CDH11-USP6) rearrangement; secondary ABC commonest with GCT. Eccentric, expansile, eggshell rim, fluid-fluid levels. Must exclude telangiectatic osteosarcoma (thick nodular septa) by biopsy. Curettage + adjuvant recurs 10-20%; polidocanol sclerotherapy is an accepted alternative.

5. UBC versus ABC versus GCT

Feature	UBC	ABC	GCT
Age	5-15	10-20	20-40 (mature)
Position	Central metaphysis	Eccentric metaphysis	Eccentric epiphyseo-metaphyseal to subchondral bone
Expansion	Not wider than physis	Blow-out, wider than physis	Expanded, thin cortex
Margin	Sclerotic, well-defined	Thin eggshell rim	Non-sclerotic, geographic
MRI	Single fluid cavity	Multiple fluid-fluid levels	Solid, low T2 areas
Genetics	NFATC2 fusions (recent)	USP6 rearrangement	H3F3A G34W
Key sign	Fallen fragment	Fluid-fluid levels	Reaches subchondral bone

6. Fibrous and other cystic-looking lesions (common distractors)

Non-ossifying fibroma (NOF) and fibrous cortical defect

- **Commonest bone lesion in children** (fibrous cortical defects in about 30% of children); age 5-20; **"don't touch" lesion**.
- **Distal femur and distal or proximal tibia metaphysis**; **eccentric, cortically based, multiloculated lucency with a scalloped sclerotic margin**; long axis parallel to the bone; migrates away from the physis and **heals spontaneously by sclerosis** in adolescence.
- Histology: storiform fibroblasts, giant cells, haemosiderin and foam cells; **KRAS / FGFR1 (MAPK pathway)** mutations.
- **Fracture risk (Arata):** lesion involving **over 50% of the transverse diameter on AP and lateral views and over 33 mm long**.
- **Treatment:** observation; pathological fractures usually heal in a cast; curettage and grafting for large symptomatic lesions or after fracture with persistent risk.
- **Jaffe-Campanacci syndrome:** multiple NOFs + café-au-lait spots (+ learning difficulty, hypogonadism, ocular and cardiac anomalies); now considered part of **neurofibromatosis type 1** spectrum.

Fibrous dysplasia (FD)

- **Activating GNAS mutation (Gs alpha, codon R201)**, post-zygotic somatic mosaic - **not inherited**; increased cAMP leads to immature fibro-osseous tissue.
- **Monostotic (about 70-80%): ribs (commonest), proximal femur, craniofacial bones**, tibia. **Polyostotic:** often unilateral, femur, tibia, pelvis.
- **McCune-Albright syndrome:** polyostotic FD + **café-au-lait spots with irregular "coast of Maine" borders** (NF1 spots have smooth "coast of California" borders) + **endocrinopathy** (**precocious puberty** commonest, especially girls; hyperthyroidism, growth hormone excess, Cushing syndrome) and **FGF23-mediated renal phosphate wasting (hypophosphataemic rickets/osteomalacia)**.
- **Mazabraud syndrome:** FD + **intramuscular myxomas**.
- **Radiograph: ground-glass** matrix, well-defined lesion with a **thick sclerotic "rind"**, expansion, endosteal scalloping; long lesion in the diaphysis/metaphysis; **shepherd's crook deformity** (coxa vara of the proximal femur); bowing; leg-length discrepancy.

- **Histology:** irregular curvilinear woven bone trabeculae (**"Chinese letters" / "alphabet soup"**) in fibrous stroma, **without osteoblastic rimming**.
- **Management:**
  - Observation for asymptomatic lesions; **bisphosphonates** (IV pamidronate/zoledronate) reduce pain (do not heal the lesion).
  - **Treat hypophosphataemia** and endocrine disease before surgery.
  - **Autogenous cancellous graft is resorbed and replaced by dysplastic bone**; use **cortical allograft struts** (resorb slowly) if grafting.
  - **Proximal femur / shepherd's crook: valgus osteotomy with intramedullary fixation (cephalomedullary nail)** - load-sharing intramedullary devices are preferred over plates in children and for long-term protection.
  - Malignant transformation (osteosarcoma, fibrosarcoma) **under 1%**, higher in McCune-Albright and after **radiotherapy (avoid)**.

Osteofibrous dysplasia (OFD, Campanacci lesion) and adamantinoma

Feature	Osteofibrous dysplasia	Adamantinoma
Age	**Under 10 years**	20-40 years
Site	**Anterior cortex of the tibial diaphysis** (also fibula); anterior bowing	**Anterior tibial diaphysis** (about 85-90% in the tibia)
Radiograph	Intracortical multiloculated lucencies with sclerosis	Multiloculated "soap-bubble" lytic lesion, may breach cortex, satellite lesions
Histology	Woven bone **with osteoblastic rimming** (unlike FD), scattered keratin-positive cells	**Epithelial (keratin-positive) nests** in fibro-osseous stroma
Nature	Benign; may regress after puberty	**Low-grade malignant**; late lung and node metastases (about 15-30%)
Treatment	**Observation**; bracing for bowing; delay surgery until after about age 15 (early resection recurs)	**Wide en bloc resection** with intercalary reconstruction (allograft, vascularised fibula, bone transport) or amputation; chemo- and radio-resistant

- **OFD-like (differentiated) adamantinoma** is an intermediate entity; OFD and adamantinoma are thought to form a spectrum, so biopsy aggressive or enlarging OFD-like lesions.

Langerhans cell histiocytosis (eosinophilic granuloma)

- Clonal proliferation of Langerhans-type dendritic cells; **BRAF V600E mutation in about 50%**.
- **Age 5-15 (most under 30)**; **"great mimicker"** (of Ewing sarcoma and osteomyelitis).
- **Skull is the commonest site** (**punched-out lesion with bevelled edges**, "hole within a hole", button sequestrum), **vertebra plana** (thoracic spine, collapse with preserved discs; Calvé disease), ribs, pelvis, long bone diaphysis (may show periosteal reaction).
- **Histology:** Langerhans cells with **grooved "coffee-bean" nuclei**, eosinophils; **CD1a, CD207 (langerin) and S100 positive**; **Birbeck granules ("tennis racquet")** on electron microscopy.
- **Clinical forms:** unifocal bone (eosinophilic granuloma, commonest, best prognosis); **Hand-Schüller-Christian** (multifocal: **skull lesions, exophthalmos, diabetes insipidus**); **Letterer-Siwe** (disseminated multisystem disease under 2-3 years, poor prognosis).
- **Treatment:** skeletal survey/PET to assess spread; unifocal lesions often **heal spontaneously** - biopsy, then observation, **intralesional methylprednisolone**, or curettage; **vertebra plana**: observation ± brace (partial height restoration in children), surgery only for neurological deficit or instability; low-dose radiotherapy rarely; **multisystem disease: chemotherapy (vinblastine + prednisolone)**, BRAF inhibitors in refractory cases.

Other lucent lesions

- **Brown tumour (osteitis fibrosa cystica):** hyperparathyroidism; multiple lytic lesions, subperiosteal resorption of the **radial side of middle phalanges**, salt-and-pepper skull; giant cells on histology mimic GCT; **resolve after parathyroidectomy**.
- **Intraosseous ganglion:** subchondral, adults, medial malleolus, around the knee and carpal bones; sclerotic rim; curettage if symptomatic.

- **Epidermoid inclusion cyst: distal phalanx** or skull after penetrating trauma; well-defined lytic lesion; curettage.
- **Intraosseous lipoma:** calcaneus (central calcification), proximal femur; observation.
- **Mnemonic for lucent lesions (FEGNOMASHIC):** Fibrous dysplasia, Enchondroma / Eosinophilic granuloma, GCT, NOF, Osteoblastoma, Metastasis / Myeloma, ABC, Simple bone cyst, Hyperparathyroidism (brown tumour), Infection, Chondroblastoma / Chondromyxoid fibroma. Lesions not likely over 30 years: EG, NOF, ABC, SBC, chondroblastoma; GCT needs a closed physis.

Small round cell tumours

1. Overview: the small round blue cell tumours

- On H&E these tumours look alike: sheets of **small, uniform cells with round nuclei and scant cytoplasm** ("small round blue cells"). Diagnosis depends on **immunohistochemistry, cytogenetics/molecular tests and age**, so biopsy tissue must be sent **fresh** (for FISH, RT-PCR, next-generation sequencing) as well as in formalin.
- **Members examined in orthopaedics:**
  - **Ewing sarcoma** (bone and soft tissue; includes the old "PNET" and **Askin tumour** of the chest wall)
  - **Primary lymphoma of bone**
  - **Multiple myeloma and solitary plasmacytoma**
  - **Metastatic neuroblastoma** (young children)
  - **Rhabdomyosarcoma** (alveolar type)
  - **Leukaemia** deposits
  - Others: **small cell osteosarcoma**, **mesenchymal chondrosarcoma**, **Langerhans cell histiocytosis** (covered with cystic lesions), osteomyelitis (inflammatory cells can mimic it).
- Mnemonic **LEMON** (Lymphoma, Ewing, Myeloma, Osteomyelitis/LCH, Neuroblastoma) is a common way to recall the differential for a permeative lesion.

Age as the first clue (classic exam rule)

Age	Most likely small round cell lesion of bone
Under 5 years	**Metastatic neuroblastoma**, leukaemia, LCH
5-30 years	**Ewing sarcoma** (peak 10-20 years)
Over 30-40 years	**Lymphoma** (adults of any age), **myeloma** (over 40-50), metastatic small cell lung carcinoma

Immunohistochemistry and genetics

Tumour	Markers	Genetic hallmark
Ewing sarcoma	**CD99 (MIC2) strong membranous**, **NKX2.2**, FLI1 nuclear; PAS positive (glycogen)	**t(11;22)(q24;q12) EWSR1-FLI1** (about 85%); t(21;22) EWSR1-ERG (about 10%)
Lymphoma (DLBCL)	**CD45 (leucocyte common antigen)**, **CD20** (B cell)	Variable
Myeloma / plasmacytoma	**CD138**, CD38, kappa or lambda light-chain restriction	t(4;14), t(14;16), del(17p) = high risk
Neuroblastoma	Synaptophysin, chromogranin, NSE; Homer Wright rosettes	**MYCN amplification** (poor prognosis)
Rhabdomyosarcoma	**Desmin, myogenin, MyoD1**	Alveolar: **t(2;13) PAX3-FOXO1**, t(1;13) PAX7-FOXO1
Mesenchymal chondrosarcoma	CD99 may be positive (pitfall), SOX9; islands of cartilage	**HEY1-NCOA2**
Small cell osteosarcoma	CD99 may be positive; **osteoid** production is the key	Not EWSR1-rearranged
Desmoplastic small round cell tumour	Desmin (dot-like) plus keratin	t(11;22)(p13;q12) EWSR1-WT1

- **WHO 2020** separates **"Ewing-like" round cell sarcomas** from true Ewing sarcoma: **CIC-DUX4 sarcoma**, **BCOR-CCNB3 sarcoma** and sarcomas with EWSR1 fused to non-ETS partners. They are mostly soft-tissue tumours, may be weakly CD99 positive, and CIC-rearranged sarcomas behave more aggressively.

> Exam pearl: CD99 is sensitive but not specific for Ewing sarcoma (also lymphoblastic lymphoma, synovial sarcoma, mesenchymal chondrosarcoma). The confirmatory test is demonstration of an EWSR1 rearrangement (FISH) or the EWSR1-FLI1 fusion transcript (RT-PCR), t(11;22) being the commonest.

2. Ewing sarcoma

Epidemiology

- **Second commonest primary malignant bone tumour in children and adolescents** (after osteosarcoma); overall about 3rd-4th commonest primary bone malignancy if myeloma is counted.
- **Age 5-25 years, peak in the second decade (10-15 years)**; rare under 5 and over 30.
- **Slight male predominance**. Much **commoner in people of European ancestry**, **rare in people of African ancestry** and less common in East Asians (germline polymorphisms near EGR2 that interact with the fusion protein).
- **Not hereditary**; no association with radiation or prior lesions (unlike osteosarcoma).
- **Cell of origin:** thought to be a mesenchymal stem cell; the EWSR1-FLI1 fusion acts as an aberrant transcription factor.

Sites

- **Long bones about 50%**: **femur is the commonest single bone**, then tibia, humerus, fibula. Classically **diaphyseal or metadiaphyseal** (contrast osteosarcoma: metaphyseal).
- **Flat bones about 40-50%**: **pelvis** (ilium most), ribs, scapula, sacrum, vertebrae. **Pelvis plus femur** account for about half of all cases.
- **Extraskeletal Ewing sarcoma** (about 15-20%): soft tissues of trunk and limbs; behaves similarly and is treated the same way.
- **Askin tumour:** Ewing sarcoma of the **thoracoabdominal wall** (chest wall, ribs).

Clinical features

- **Pain** (often intermittent at first, worse at night), **swelling**, palpable tender mass, warmth.
- **Systemic features in 10-20%: fever, weight loss, anaemia, raised ESR, raised WBC and raised LDH**, which makes it the **classic mimic of osteomyelitis**. Biopsy may yield "pus-like" liquefied necrotic tumour.
- Pathological fracture in about 10-15%.

- Rib or pelvic tumours present late with a large mass, pleural effusion or sciatica-like pain; spinal or sacral tumours with radicular symptoms or cord compression.

Radiology

- **Plain radiograph:**
  - **Permeative or moth-eaten lysis** (wide zone of transition) in the diaphysis.
  - **Lamellated "onion-skin" periosteal reaction** (repeated cycles of periosteal elevation).
  - **Codman triangle**; **"hair-on-end"** or sunburst spiculation in some.
  - **Large soft-tissue mass out of proportion to the bone destruction**.
  - **Saucerisation** (external erosion of the cortex by a subperiosteal mass) in periosteal-located tumours.
  - Sclerotic (reactive) changes in up to a third, especially pelvic tumours: do not assume osteosarcoma.
- **MRI of the whole bone** (with adjacent joints): extent of marrow and soft-tissue disease, **skip lesions**, neurovascular relations; plan biopsy and resection. The soft-tissue component often dwarfs the bone change.
- **Staging (systemic):**
  - **CT chest** (lungs are the commonest site of metastasis).
  - **Whole-body bone imaging: PET-CT** (now preferred; FDG-avid) or bone scintigraphy, whole-body MRI in some protocols.
  - **Bone marrow aspirate and trephine biopsy** (traditionally bilateral posterior iliac crests) because Ewing sarcoma spreads to marrow; many groups now omit it when PET-CT is negative (debated).
  - Bloods: FBC, ESR, **LDH** (prognostic), renal and liver function, fertility counselling before chemotherapy.
- **Biopsy:** core needle (image-guided) or open biopsy **planned with the definitive surgeon**, in line with the future incision, through a single compartment; soft-tissue component usually gives diagnostic tissue. Send **fresh tissue for molecular studies** and **culture** (to exclude infection).

Staging and prognosis

- Ewing sarcoma is by definition **high grade** (Enneking stage IIB if localised and extracompartmental, III if metastatic; AJCC staging also used).
- **About 20-25% have detectable metastases at diagnosis**: **lungs** (commonest), **bone and bone marrow**. Micrometastatic disease is assumed in all, hence chemotherapy for everyone.
- **Survival (approximate):** localised disease **about 70-75% at 5 years**; isolated lung metastases about 40%; bone or marrow metastases under 20-30%; relapse under 15%.

Prognostic factor	Unfavourable
Metastases at diagnosis	**Most important adverse factor**; bone/marrow worse than lung only
Site	**Pelvis, axial and proximal** sites worse than distal limb
Tumour size/volume	Over 8 cm or **over 200 mL**
Age	Older (over 15-18 years, adults) worse
Serum LDH	Raised
Histological response to induction chemotherapy	Poor response (**viable tumour 10% or more**, i.e. under 90% necrosis)
Fever, anaemia at presentation	Worse
Fusion type	Not prognostic with modern therapy (EWSR1-FLI1 vs ERG)

Management (multimodal; treat in a sarcoma centre)

1. **Neoadjuvant (induction) chemotherapy** for all patients after biopsy and staging.
2. **Local control** (surgery and/or radiotherapy) at about week 12-18, after reassessment imaging.
3. **Adjuvant (consolidation) chemotherapy**; total treatment about **9-12 months**.

Chemotherapy

- Backbone drugs: **vincristine, doxorubicin, cyclophosphamide, ifosfamide, etoposide** (actinomycin D in some regimens).

- **North American standard: VDC alternating with IE**. **COG AEWS0031 (Womer et al., 2012):** giving VDC/IE **every 2 weeks (interval-compressed)** instead of every 3 weeks **improved event-free survival** without more toxicity.
- **Euro Ewing 2012 trial (Brennan et al., Lancet 2022): interval-compressed VDC/IE was superior to the European VIDE induction regimen** (better event-free and overall survival, less toxicity). VDC/IE is now the standard on both sides of the Atlantic.
- **High-dose busulfan-melphalan with autologous stem-cell rescue:** Euro-EWING 99/R2Loc (Whelan et al., 2018) showed better event-free survival than standard chemotherapy in **high-risk localised disease** (poor histological response or large tumour volume); no benefit in pulmonary metastases (R2Pulm).
- Late effects: cardiomyopathy (doxorubicin), infertility and haemorrhagic cystitis (alkylating agents), nephrotoxicity (ifosfamide), **secondary leukaemia/MDS** (etoposide, alkylators).

Local control: surgery versus radiotherapy

- **Ewing sarcoma is radiosensitive** (unlike osteosarcoma), so local control can be by surgery, radiotherapy or both. No randomised trial compares them; retrospective data favour **surgery when a wide margin can be achieved with acceptable function** (lower local recurrence), partly due to selection bias.

Option	Indications	Notes
Surgery alone (wide resection)	Resectable tumour, wide margins achievable, **good histological response**	**Preferred where feasible**; avoids radiation late effects. Reconstruction: endoprosthesis, allograft, vascularised fibula, rotationplasty, expendable bones (fibula, rib, clavicle) need no reconstruction

Option	Indications	Notes
Surgery plus postoperative radiotherapy	**Inadequate (marginal/intralesional) margins**, **poor histological response** (10% or more viable cells), some pelvic tumours	Doses around 45-55 Gy
Definitive radiotherapy alone	**Unresectable or functionally unacceptable resection**: spine, sacrum, many pelvic and skull-base tumours	About 55-56 Gy to the tumour; higher local recurrence than surgery in most series
Whole-lung irradiation	Lung metastases	Improves outcome in pulmonary metastatic disease

- **Radiotherapy complications: radiation-induced sarcoma** (commonly osteosarcoma, latency often over 5-10 years, dose-related), **physeal arrest and limb-length discrepancy**, **pathological fracture**, joint contracture, soft-tissue fibrosis, infertility (pelvic fields). This is why surgery is preferred in growing children when feasible.
- **Pathological fracture at presentation:** still treat with neoadjuvant chemotherapy first; splint or cast rather than internal fixation (fixation contaminates the whole bone); limb salvage remains possible.
- **Relapse:** early (under 2 years) relapse has a very poor prognosis; options include irinotecan-temozolomide, high-dose ifosfamide, high-dose chemotherapy with stem-cell rescue, and surgery or radiotherapy of oligometastases.

Differential diagnosis (distractors)

Condition	Distinguishing features
Acute or subacute osteomyelitis	Systemic illness more acute; MRI shows abscess, sequestrum; cultures positive; responds to antibiotics. **Always send biopsy for both culture and histology**
Langerhans cell histiocytosis	Age under 10, punched-out lesion, "vertebra plana", CD1a and langerin (CD207) positive
Osteosarcoma	Metaphyseal, **tumour osteoid** (cloud-like bone formation), raised ALP

Condition	Distinguishing features
Lymphoma of bone	Adults, CD45/CD20 positive, patient often well
Metastatic neuroblastoma	Under 5 years, raised urinary VMA/HVA, adrenal mass, MIBG avid
Stress fracture	Linear callus, history of activity, no soft-tissue mass

Exam pearl: Ewing sarcoma = second decade, diaphysis of femur or pelvis, permeative lysis, onion-skin periosteal reaction, big soft-tissue mass, fever and raised ESR/LDH (mimics osteomyelitis), CD99 positive, t(11;22) EWSR1-FLI1. Worst prognostic factor = metastases at diagnosis. Standard chemotherapy = interval-compressed VDC/IE. Radiosensitive, but surgery is preferred for local control when a wide margin is achievable.

3. Primary lymphoma of bone

Definition and epidemiology

- **Primary bone lymphoma (PBL):** lymphoma arising in bone **without nodal or visceral disease** (classically no other site within 6 months of diagnosis). About **5% of extranodal lymphomas** and about **5-7% of primary malignant bone tumours**.
- **Secondary bone involvement** by systemic lymphoma (stage IV) is much commoner than primary bone lymphoma.
- **Histology: diffuse large B-cell lymphoma (DLBCL) in about 80% or more**; others follicular, anaplastic large cell, Hodgkin (rare in bone).
- **Adults, peak 40-60 years** (wide range, can occur in children), **male predominance**.
- **Sites: femur (commonest)**, pelvis, humerus, tibia, spine; metadiaphyseal.

Clinical and radiological features

- Bone pain, swelling, pathological fracture (about 20%), cord compression in spinal disease. **B symptoms (fever, night sweats, weight loss) uncommon**; the patient often looks **well relative to the size of the lesion**.
- **Radiographs: permeative or moth-eaten lysis** with a wide zone of transition, **little or no periosteal reaction**, sometimes mixed lytic-sclerotic;

radiographs can look **nearly normal despite extensive marrow involvement** (cortex permeated by small channels rather than destroyed).

- **MRI:** extensive marrow replacement with a **large circumferential soft-tissue mass** around a relatively intact cortex.
- **Ivory vertebra:** Hodgkin lymphoma (differentials: osteoblastic prostate or breast metastasis, Paget disease).
- **Staging: PET-CT** (Lugano staging), bone marrow biopsy, CT chest/abdomen/pelvis, LDH.

Management

- **Biopsy** (core, with fresh tissue for flow cytometry); the diagnosis is made on CD20/CD45 positivity.
- **Chemotherapy: R-CHOP** (rituximab, cyclophosphamide, doxorubicin, vincristine, prednisolone) for DLBCL, often followed by **consolidation radiotherapy** to the involved bone (benefit of adding radiotherapy is debated).
- **Surgery has no role in tumour removal.** Orthopaedic roles: **biopsy**, **fixation of pathological or impending fractures** (intramedullary nailing, cemented arthroplasty), decompression for cord compression when unstable.
- **Prognosis is good**: 5-year survival about 70-90% for localised DLBCL of bone, better than for most other primary bone malignancies.

> Exam pearl: An adult with a permeative lytic lesion, a large soft-tissue mass, minimal periosteal reaction and surprisingly few symptoms suggests primary lymphoma of bone. It is CD45/CD20 positive DLBCL, and it is treated with chemotherapy (R-CHOP) with or without radiotherapy, not resection.

4. Multiple myeloma and plasmacytoma

Epidemiology and pathology

- **Commonest primary malignant tumour arising in bone** (when included among bone tumours); bone involvement in about 80% at diagnosis.

- **Age over 40, median about 65-70 years**; **male predominance**; about **twice as common in people of African ancestry**.
- Clonal proliferation of **plasma cells** secreting a **monoclonal immunoglobulin (M-protein)**: **IgG commonest (about 50%)**, then **IgA (about 20%)**, light-chain-only (about 20%); IgD and non-secretory rare.
- Precursors: **MGUS** (progression about **1% per year**) and smouldering myeloma.
- **Bone destruction:** myeloma cells increase **RANKL** and reduce OPG, activating **osteoclasts**, and secrete **DKK1** (and sclerostin), which **suppresses osteoblasts**. Hence **purely lytic lesions, normal alkaline phosphatase, and a "cold" bone scan**.
- **Histology:** sheets of plasma cells (eccentric "clock-face" nucleus, perinuclear hof), **CD138 and CD38 positive**, kappa or lambda restricted. Amyloid may be present.

Clinical features (CRAB)

- **C**alcium raised (hypercalcaemia: confusion, constipation, polyuria).
- **R**enal impairment (light-chain cast nephropathy, hypercalcaemia, amyloid).
- **A**naemia (normocytic; marrow replacement): fatigue.
- **B**one disease: **back pain (commonest presenting symptom)**, vertebral collapse, pathological fractures, cord compression.
- Others: recurrent infections (immunoparesis), hyperviscosity, **amyloidosis** (carpal tunnel syndrome, macroglossia), peripheral neuropathy.

Diagnostic criteria (IMWG 2014)

- **Clonal bone marrow plasma cells 10% or more, or biopsy-proven bony or extramedullary plasmacytoma**, PLUS **one or more myeloma-defining events**:
  - **CRAB** features attributable to the plasma-cell disorder: calcium over 2.75 mmol/L (11 mg/dL), creatinine over 177 micromol/L (2 mg/dL) or clearance under 40 mL/min, haemoglobin under 10 g/dL (or 2 g/dL below normal), **one or more osteolytic lesions** (on CT, low-dose whole-body CT or PET-CT).
  - **SLiM biomarkers**: **Sixty** percent or more clonal plasma cells; **Light-chain** ratio (involved:uninvolved serum free light chain) 100 or more; **MRI** with

more than one focal lesion of 5 mm or more.

Investigations

- **Bloods:** FBC, **ESR very high**, **rouleaux** on smear, calcium, creatinine, albumin, **beta-2 microglobulin**, LDH, **alkaline phosphatase normal**.
- **Serum protein electrophoresis with immunofixation** (M band), **serum free light-chain assay**, **urine for Bence Jones protein** (light chains; not detected by dipstick).
- **Bone marrow aspirate and trephine**, with cytogenetics/FISH.
- **Imaging:**
  - **Whole-body low-dose CT** is now preferred over the conventional skeletal survey (IMWG); **PET-CT** and **whole-body MRI** (spine and pelvis; detects marrow infiltration and SLiM focal lesions).
  - **Bone scan is unreliable (false negative)** because there is little osteoblastic activity: in suspected myeloma use radiographs/CT/MRI, not scintigraphy.
- **Radiographic signs: "punched-out" lytic lesions without a sclerotic rim**, **"raindrop" (pepper-pot) skull**, diffuse osteopenia, vertebral collapse (**pedicles usually spared early**, as the pedicle has less red marrow; contrast metastases, the "winking owl" sign), expansile lytic lesion in a rib or pelvis. **Sclerotic myeloma** occurs in **POEMS syndrome** (polyneuropathy, organomegaly, endocrinopathy, M-protein, skin changes).

Staging

System	Basis
ISS (International Staging System)	**Stage I:** beta-2 microglobulin under 3.5 mg/L and albumin 3.5 g/dL or more. **Stage III:** beta-2 microglobulin 5.5 mg/L or more. **Stage II:** neither I nor III
R-ISS (Revised ISS)	Adds **LDH** and **high-risk cytogenetics** (del(17p), t(4;14), t(14;16))
Durie-Salmon	Older system based on Hb, calcium, M-protein level and number of bone lesions

Management

- **Systemic therapy (haematology):** triplet or quadruplet regimens, e.g. **bortezomib (proteasome inhibitor), lenalidomide (immunomodulator), dexamethasone (VRd)**, often with the anti-CD38 antibody **daratumumab**; **autologous stem-cell transplantation** in fit patients; newer CAR-T and bispecific antibodies for relapse.
- **Bone-targeted therapy: zoledronic acid** (reduces skeletal events and has a modest survival benefit, MRC Myeloma IX) or **denosumab** (useful in renal impairment); calcium and vitamin D; watch for **osteonecrosis of the jaw**.
- **Avoid NSAIDs and nephrotoxic contrast**; hydrate.
- **Radiotherapy:** pain control, impending fracture, cord compression (myeloma is **very radiosensitive**; low doses, e.g. 8 Gy single fraction or 20 Gy in 5 fractions).
- **Orthopaedic and spinal surgery:**
  - **Biopsy any solitary lytic lesion** before fixation if the diagnosis is not established (a solitary lytic lesion may be myeloma, renal metastasis or primary sarcoma).
  - **Prophylactic fixation of impending fractures** and fixation of pathological fractures: **intramedullary nailing protecting the whole bone**, cemented arthroplasty for the femoral neck, cement augmentation. **Pathological fractures in myeloma have the highest healing rate** of all bone metastatic lesions (about two-thirds heal; Gainor and Buchert).
  - **Vertebral compression fractures: kyphoplasty or vertebroplasty** for painful fractures (**CAFE trial**, Berenson 2011: kyphoplasty improved pain and function in cancer-related fractures, mostly myeloma); brace.
  - **Cord compression:** dexamethasone and radiotherapy are first-line (radiosensitive); surgery for spinal instability, bony retropulsion or neurological deterioration despite radiotherapy.

Solitary plasmacytoma

- **Solitary plasmacytoma of bone:** a single lytic lesion of clonal plasma cells with **no CRAB features** and normal (or under 10% clonal) marrow.
- **Commonest in the spine (vertebral body)**, then pelvis, femur, ribs; younger than myeloma (median about 55).

- **Treatment: radiotherapy about 40-50 Gy** (local control over 85%); surgery for instability or fracture.
- **About 50-70% progress to multiple myeloma within 10 years**: follow with SPEP and free light chains.
- **Extramedullary plasmacytoma** (soft tissue, upper aerodynamic tract) has a lower rate of progression.

> Exam pearl: Myeloma lesions are purely lytic with normal ALP and a falsely negative bone scan, because osteoblasts are suppressed. Image with whole-body low-dose CT, PET-CT or MRI. A solitary lytic lesion in a patient over 40 must be biopsied before fixation.

5. Metastatic neuroblastoma

- **Commonest extracranial solid tumour of childhood**; **most present under 5 years (median about 18 months)**; arises from neural crest cells of the **adrenal medulla** or sympathetic chain.
- **Commonest cause of bone metastases in children under 5**; metastases to **skull and orbit (periorbital bruising, "raccoon eyes", proptosis)**, long-bone metaphyses (lytic, permeative, symmetric lesions), marrow ("Hutchinson" type spread to bone).
- **Clinical:** abdominal mass crossing the midline, limp, bone pain, irritability, fever (mimics osteomyelitis or leukaemia); opsoclonus-myoclonus syndrome; **Horner syndrome** (cervical/thoracic).
- **Investigations: raised urinary catecholamine metabolites (VMA and HVA)**; **MIBG scintigraphy** (iodine-123 metaiodobenzylguanidine) for primary and metastases; marrow biopsy.
- **Histology:** small round blue cells with **Homer Wright rosettes**, neuropil.
- **Prognosis:** better under 18 months; **MYCN amplification**, unfavourable histology (Shimada) and 11q loss are adverse. **Stage MS (formerly 4S)**: infants under 18 months with limited metastases (skin, liver, marrow) may regress spontaneously.
- **Spinal "dumbbell" tumour** through the neural foramen can cause cord compression and later post-laminectomy or post-radiation **kyphosis or scoliosis**.

6. Rhabdomyosarcoma and leukaemia (orthopaedic relevance)

Rhabdomyosarcoma

- **Commonest soft-tissue sarcoma of childhood**.

Type	Features
Embryonal (commonest)	Younger children (under 10), head and neck, genitourinary (**botryoid** variant: grape-like vaginal or bladder mass); better prognosis
Alveolar	**Adolescents, extremities** and trunk; **t(2;13) PAX3-FOXO1** or t(1;13) PAX7-FOXO1; worse prognosis; can metastasise to bone and marrow (mimics leukaemia) and lymph nodes
Pleomorphic	Adults; high grade

- **Treatment:** multi-agent chemotherapy (**VAC: vincristine, actinomycin D, cyclophosphamide**) plus local control by surgery and/or radiotherapy; **regional lymph-node evaluation** (sentinel node biopsy) for extremity tumours.

Acute leukaemia

- **ALL is the commonest childhood cancer**; up to about a third present with **bone pain, limp or arthralgia** and may be misdiagnosed as **juvenile idiopathic arthritis or osteomyelitis**.
- **Radiographic clues: metaphyseal lucent bands ("leukaemic lines")**, diffuse osteopenia, vertebral compression fractures, permeative lesions.
- **Red flags:** night pain, pain out of proportion to joint signs, cytopenias, blasts on smear, raised LDH and uric acid, hepatosplenomegaly, lymphadenopathy. **Do a peripheral smear before starting steroids for "JIA".**

7. Comparison of the main small round cell tumours of bone

Feature	Ewing sarcoma	Lymphoma of bone	Myeloma	Neuroblastoma metastasis
Age	5-25 (peak 10-15)	40-60 (any)	Over 40 (median 65-70)	Under 5

Feature	Ewing sarcoma	Lymphoma of bone	Myeloma	Neuroblastoma metastasis
Site	Femur diaphysis, pelvis, ribs	Femur, pelvis, spine	Spine, skull, ribs, pelvis, proximal long bones	Skull, orbit, metaphyses
Radiograph	Permeative lysis, onion skin, big soft-tissue mass	Permeative, minimal periosteal reaction, big soft-tissue mass	Punched-out lysis, no sclerotic rim	Symmetric lytic metaphyseal lesions
Key test	CD99, EWSR1-FLI1	CD45, CD20	M-protein, CD138, marrow	Urine VMA/HVA, MIBG
Bone scan	Hot	Hot	**Often cold**	Hot (MIBG better)
Mainstay of treatment	Chemo plus surgery and/or RT	Chemo (R-CHOP) plus or minus RT	Systemic therapy, bisphosphonate, RT	Chemo, surgery, RT, immunotherapy

Exam pearl: Age sorts the round cell tumours: neuroblastoma under 5, Ewing 5-25, lymphoma in middle age, myeloma after 40-50. Radiosensitive lesions in this group (Ewing, lymphoma, myeloma, plasmacytoma) explain why radiotherapy has a much larger role here than in osteosarcoma or chondrosarcoma.

Soft-tissue sarcoma

1. Epidemiology and general features

- Soft-tissue sarcomas (STS) are malignant tumours of **mesenchymal origin** (fat, muscle, fibrous tissue, vessels, nerve sheath); **about 1% of adult cancers** but over **70 histological subtypes** (WHO 2020).
- **Benign soft-tissue tumours outnumber sarcomas by about 100:1**; **lipoma** is the commonest soft-tissue tumour.
- **Age:** any; most in adults over 50. In **children the commonest STS is rhabdomyosarcoma**.
- **Commonest STS in adults: undifferentiated pleomorphic sarcoma (UPS)** and **liposarcoma** are the two commonest (which is first varies by series; for extremity sarcomas in older adults UPS is often quoted, for all sites liposarcoma).
- **Site: extremities about 50-60%** (**lower limb, especially the thigh, is the commonest site**), trunk wall, **retroperitoneum about 15%**, head and neck, viscera.
- **Behaviour:** grow centrifugally and compress surrounding tissue into a **pseudocapsule (reactive zone)** containing **satellite tumour cells**, with **skip lesions** in high-grade tumours. **Shelling out through the pseudocapsule leaves tumour behind**, which is why wide excision is needed.
- **Spread: haematogenous**, **lungs commonest (about 80% of metastases)**. Lymph-node metastasis is **uncommon (under 5%) except in specific subtypes** (see below).
- **Predisposing factors:**
  - **Li-Fraumeni syndrome (TP53)**, **neurofibromatosis type 1** (MPNST), **hereditary retinoblastoma (RB1)**, **familial adenomatous polyposis/Gardner syndrome** (desmoid tumours), Werner syndrome.
  - **Prior radiotherapy** (radiation-associated sarcoma, latency usually over 5-10 years; angiosarcoma, UPS).
  - **Chronic lymphoedema** (**Stewart-Treves syndrome**: lymphangiosarcoma/angiosarcoma after mastectomy and axillary clearance).

- Chemicals (vinyl chloride: hepatic angiosarcoma), HIV/HHV-8 (Kaposi sarcoma).

2. Clinical evaluation of a soft-tissue mass

- Typical sarcoma: **painless, enlarging mass**; pain occurs in about a third. Patients often attribute it to trauma ("haematoma" that does not resolve).
- **Red-flag features of a soft-tissue lump (suspect sarcoma):**
  - **Size over 5 cm** (golf ball)
  - **Deep to the deep fascia**
  - **Increasing in size**
  - **Painful**
  - **Recurrence after previous excision**
- **Any one** feature warrants referral and imaging; deep or over 5 cm lumps have a high risk (about 10% or more of deep lumps over 5 cm are malignant). Superficial small lumps can still be sarcoma (e.g. epithelioid, myxofibrosarcoma).
- **Examination:** size, depth, consistency, mobility, transillumination, pulsatility, neurovascular status (Tinel sign over a nerve sheath tumour), **regional lymph nodes**, skin changes.

3. Investigations

- **Ultrasound:** first-line triage of superficial lumps (UK pathway); sarcoma suspected if **heterogeneous, vascular, deep, or over 5 cm**.
- **MRI with gadolinium: investigation of choice** for local evaluation: size, compartment, depth, relation to **neurovascular bundle and bone**, peritumoural oedema, necrosis; plans biopsy and surgery.
  - **Lipoma** is homogeneous fat signal, suppresses completely on fat-saturated sequences, with only thin septa.
  - **Atypical lipomatous tumour (ALT)**: thick (over 2 mm) or nodular septa, non-fatty areas, large size, deep location.
  - **Myxoid tumours** (myxoid liposarcoma, myxofibrosarcoma, myxoma) can look **cyst-like** (high T2) but **enhance** with contrast: never call a deep "cyst" benign without contrast.

- **Plain radiographs:** bone involvement, **calcification** (synovial sarcoma about 30%, phleboliths in haemangioma, myositis ossificans zonation), extraskeletal osteosarcoma.
- **Staging:**
  - **CT chest** for all intermediate- or high-grade sarcomas (lungs are the commonest metastatic site).
  - **Myxoid (round cell) liposarcoma:** metastasises to **extrapulmonary sites** (spine, retroperitoneum, other soft tissue) and may be missed by CT and PET: add **whole-spine MRI or whole-body MRI**.
  - **Lymph-node assessment** (clinical, ultrasound, CT/PET, sentinel node biopsy in selected cases) for node-seeking subtypes.
  - **CT abdomen/pelvis** for myxoid liposarcoma, retroperitoneal tumours and leiomyosarcoma; brain MRI for **alveolar soft part sarcoma** and angiosarcoma.
  - **PET-CT:** useful for high-grade tumours, rhabdomyosarcoma, NF1 (suspected malignant change in a plexiform neurofibroma) and response assessment.

Biopsy principles (Mankin)

- **Biopsy after all imaging**, at (or after discussion with) **the specialist sarcoma centre that will perform definitive surgery**. **Mankin et al. (1982, repeated 1996)**: biopsies performed in referring hospitals had **significantly more diagnostic errors, complications and changes in treatment (including unnecessary amputation)** than those performed in treating centres.
- **Image-guided core needle biopsy** is the standard (diagnostic accuracy over 90%); open incisional biopsy if core is non-diagnostic. **Fine-needle aspiration is not adequate** for primary diagnosis (no architecture or grade).
- **Rules:**
  - **Longitudinal** incision in the limb (in line with the definitive resection), as small as possible.
  - Through a **single compartment**; avoid intermuscular planes, joints and the **neurovascular bundle**.

- Meticulous haemostasis; if a drain is used, it exits **in line with and close to the incision**.
- Biopsy tract is considered contaminated and **excised en bloc** with the tumour.
- Sample the **enhancing, viable periphery** (avoid necrotic centre); send fresh tissue for **molecular genetics** and cultures where infection is possible.

- **Excisional biopsy** is acceptable only for **small (under 3-5 cm), superficial** lesions where re-excision would be straightforward.

4. Grading and staging

FNCLCC (French) grading system: the standard histological grade

Parameter	Score
Tumour differentiation	1 = closely resembles normal adult mesenchymal tissue; 2 = histological type certain; 3 = embryonal, undifferentiated, or type uncertain
Mitotic count (per 10 high-power fields)	1 = 0-9; 2 = 10-19; 3 = 20 or more
Tumour necrosis	0 = none; 1 = under 50%; 2 = 50% or more
Total score	**2-3 = Grade 1**; **4-5 = Grade 2**; **6-8 = Grade 3**

- **Grade is the most important predictor of distant metastasis** (and overall survival); **margin status is the most important predictor of local recurrence**.

AJCC 8th edition: trunk and extremity sarcoma

T stage	Size
T1	5 cm or less
T2	Over 5 cm up to 10 cm
T3	Over 10 cm up to 15 cm
T4	Over 15 cm

Stage	Definition
IA	T1, N0, M0, grade 1 (or GX)
IB	T2-T4, N0, M0, grade 1 (or GX)
II	T1, N0, M0, grade 2-3
IIIA	T2, N0, M0, grade 2-3
IIIB	T3-T4, N0, M0, grade 2-3
IV	**Any N1 or any M1**

- **Depth (superficial vs deep) was removed from the AJCC 8th edition staging** (it was in the 7th), though it remains prognostic and is still a red flag.
- **Retroperitoneal sarcomas** have a separate AJCC staging; Enneking staging (low/high grade, intra/extracompartmental, metastasis) is still used by orthopaedic oncologists, particularly for musculoskeletal surgical planning.

5. Important subtypes (high-yield table)

Subtype	Typical patient and site	Key pathology/genetics	Behaviour and exam points
Atypical lipomatous tumour / well-differentiated liposarcoma	Older adults; **ALT in limbs**, WDL in retroperitoneum	**MDM2 and CDK4 amplification (12q13-15 ring/giant marker chromosomes)**; MDM2 FISH distinguishes from lipoma	**Does not metastasise**; local recurrence; **marginal excision acceptable in limbs**; risk of **dedifferentiation** (higher in retroperitoneum)

Subtype	Typical patient and site	Key pathology/genetics	Behaviour and exam points
Myxoid liposarcoma (including round-cell)	Young-middle-aged adults (30-50), **thigh**	**t(12;16) FUS-DDIT3** (rarely t(12;22) EWSR1-DDIT3); **"chicken-wire" capillaries**, lipoblasts	**Very radiosensitive** and chemosensitive (trabectedin); **extrapulmonary metastases** (spine, retroperitoneum, soft tissues); round-cell over 5% = higher grade
Pleomorphic and dedifferentiated liposarcoma	Older adults	Pleomorphic lipoblasts; dedifferentiated keeps MDM2 amplification	High grade
Undifferentiated pleomorphic sarcoma (UPS)	**Elderly (over 50-60), thigh**, deep	Diagnosis of exclusion; formerly "malignant fibrous histiocytoma"	High grade; lung metastases; also arises in bone (infarct, Paget, radiation)
Myxofibrosarcoma	Elderly, **superficial**, limbs	Myxoid stroma, curvilinear vessels	**Infiltrative "tail sign" along fascia on MRI**; **very high local recurrence**; needs wide margins

Subtype	Typical patient and site	Key pathology/genetics	Behaviour and exam points
Synovial sarcoma	**Adolescents and young adults (15-40)**, **around the knee (popliteal fossa)**, foot and ankle; **rarely intra-articular** despite the name	**t(X;18)(p11;q11) SS18-SSX1/SSX2**; **TLE1** and **SS18-SSX fusion-specific antibody** positive; biphasic or monophasic	Often **slow-growing and painful**, may be small and mistaken for benign; **calcification in about 30%** on radiographs; **chemosensitive (ifosfamide)**; lymph-node metastases; late metastases
Epithelioid sarcoma	**Young adults, hand, wrist and forearm (commonest STS of the hand in many series)**	**Loss of INI1 (SMARCB1)**; keratin and EMA positive	Ulcerating nodules mistaken for **granuloma, infection or rheumatoid nodule**; spreads along fascia and tendons; **lymph-node metastases**; tazemetostat (EZH2 inhibitor)
Clear cell sarcoma ("melanoma of soft parts")	Young adults, **foot and ankle tendons and aponeuroses**	**t(12;22) EWSR1-ATF1**; **S100, HMB-45, Melan-A positive** (melanocytic markers)	Lymph-node metastases; poor prognosis; distinguished from melanoma by the fusion

Subtype	Typical patient and site	Key pathology/genetics	Behaviour and exam points
Alveolar soft part sarcoma	Adolescents and young adults, thigh/buttock	**t(X;17) ASPSCR1-TFE3**; TFE3 nuclear	Highly vascular; **metastases to lung and brain** early; responds to immune checkpoint inhibitors
Malignant peripheral nerve sheath tumour (MPNST)	About half in **NF1** (lifetime risk about 8-13%), sciatic nerve, brachial plexus	S100 patchy or lost, loss of H3K27me3	Suspect when a **plexiform neurofibroma becomes painful or grows rapidly, or a new neurological deficit appears**; FDG-PET (high SUV) guides biopsy; poor prognosis
Leiomyosarcoma	Middle-aged and elderly; **retroperitoneum, IVC**, uterus, limbs (vascular origin)	Desmin, SMA, h-caldesmon	Haematogenous spread
Angiosarcoma	Elderly scalp, post-radiation breast, **Stewart-Treves** (lymphoedematous arm)	CD31, ERG, CD34	Very aggressive, multifocal
Dermatofibrosarcoma protuberans (DFSP)	Young-middle-aged adults, trunk, superficial	**COL1A1-PDGFB**, CD34 positive	Locally aggressive, rarely metastasises; wide excision or **Mohs**; **imatinib** for unresectable

Subtype	Typical patient and site	Key pathology/genetics	Behaviour and exam points
Rhabdomyosarcoma	Children; alveolar type in adolescent extremities	Desmin, myogenin; PAX3/7-FOXO1	VAC chemotherapy; lymph nodes
Extraskeletal myxoid chondrosarcoma	Adults, thigh	NR4A3 rearrangement (EWSR1-NR4A3)	Slow but metastasises

Lymph-node-seeking sarcomas (classic list, mnemonic "SCARE")

- **S**ynovial sarcoma, **C**lear cell sarcoma, **A**ngiosarcoma, **R**habdomyosarcoma, **E**pithelioid sarcoma (alveolar soft part sarcoma also).

> Exam pearl: Lymph node spread is rare in sarcoma except SCARE (synovial, clear cell, angiosarcoma, rhabdomyosarcoma, epithelioid). Myxoid liposarcoma spreads to extrapulmonary sites, so stage with whole-spine or whole-body MRI. Grade predicts metastasis; margin predicts local recurrence.

6. Intermediate (locally aggressive) tumours: desmoid fibromatosis

- **Desmoid-type (aggressive) fibromatosis:** clonal myofibroblastic proliferation; **locally infiltrative, never metastasises**, but recurs often and can cause morbidity.
- **Genetics:** sporadic tumours have **CTNNB1 (beta-catenin) mutations** (S45F mutation associated with higher recurrence); **APC germline mutation** in **familial adenomatous polyposis/Gardner syndrome** (intra-abdominal, mesenteric desmoids). **Nuclear beta-catenin** on immunohistochemistry.
- Sites: shoulder girdle, chest wall, thigh, abdominal wall (often **postpartum**; oestrogen influence), mesentery.
- **Natural history is unpredictable**: a substantial proportion stabilise or **regress spontaneously**.
- **Management (Desmoid Tumor Working Group global consensus 2020):**
  - **Active surveillance first-line** for most asymptomatic or non-threatening tumours, with MRI after 1-2 months and then every 3-6 months.

- Treat only if **persistent progression** or symptoms, or threat to vital structures.
- **Systemic therapy is preferred over surgery** for progressive disease: **nirogacestat** (gamma-secretase inhibitor; **DeFi trial**, NEJM 2023, improved progression-free survival), **sorafenib** (Gounder et al., NEJM 2018), low-dose methotrexate-vinblastine, pegylated liposomal doxorubicin, NSAIDs/hormonal agents (weak evidence).
- **Surgery** (aiming for negative margins without major functional loss) for selected progressive limb or abdominal-wall tumours; **positive margins are often watched** rather than re-excised. Radiotherapy, cryoablation are alternatives.

7. Management of soft-tissue sarcoma

Surgery: wide local excision is the mainstay

- **Margins (Enneking):**

Margin	Plane	Local recurrence (without radiotherapy)
Intralesional	Through the tumour	Almost certain
Marginal	Through the **reactive zone/pseudocapsule**	High for high-grade tumours (satellite cells)
Wide	Through normal tissue with a **cuff of normal tissue** around the reactive zone	Acceptable, especially combined with radiotherapy
Radical	**Entire compartment** removed (or amputation proximal to the compartment)	Lowest; rarely needed now

- In practice, a **wide margin of about 1-2 cm of normal tissue, or an intact fascial barrier** (fascia, periosteum, epineurium, vessel adventitia act as barriers), is aimed for. **R0** = microscopically negative, **R1** = microscopic tumour at margin, **R2** = macroscopic tumour left.
- **Limb salvage is possible in over 90% of extremity sarcomas**. **Rosenberg et al. (NCI, 1982): limb-sparing surgery plus radiotherapy gave the same**

overall survival as amputation (higher local recurrence, but no survival difference). This established limb salvage.

- **Amputation** indications: tumour encasing major neurovascular structures where reconstruction is impossible or would leave a useless limb, extensive contamination (failed previous surgery, fungating tumour), recurrence after maximum radiotherapy, patient choice.
- Neurovascular involvement: **vascular resection and reconstruction** (graft) is possible; **epineurial dissection** if the nerve is abutted, nerve resection if encased (sciatic resection may still give a functional limb).
- Soft-tissue reconstruction: local or **free flaps** (especially after preoperative radiotherapy to reduce wound complications).

Unplanned excision ("whoops procedure")

- Excision of a mass without preoperative imaging and assumed to be benign, later found to be a sarcoma; usually **intralesional or marginal**.
- **Residual tumour is found in about 40-50% of re-excision specimens**.
- Management: refer to a sarcoma centre, **MRI (and staging)**, then **re-excision of the entire tumour bed and scar with wide margins**, usually with radiotherapy. Prognosis after appropriate re-excision approaches that of planned surgery, but morbidity (larger resections, flaps) is higher.

Radiotherapy

- **Indications:** most **intermediate- and high-grade, deep or over 5 cm** tumours, and **close or positive margins**. Low-grade, small, superficial tumours excised with wide margins may need surgery alone.
- **Yang et al. (NCI, 1998)** randomised trial: adjuvant external-beam radiotherapy after limb-sparing surgery **reduced local recurrence but did not change overall survival**.
- **Preoperative versus postoperative radiotherapy: O'Sullivan et al. (NCIC SR2, Lancet 2002):**

Feature	Preoperative RT (50 Gy)	Postoperative RT (60-66 Gy)
Major wound complications	**Higher (about 35%)**	Lower (about 17%)

Feature	Preoperative RT (50 Gy)	Postoperative RT (60-66 Gy)
Late toxicity (fibrosis, joint stiffness, oedema)	**Lower** (Davis 2005)	**Higher**
Dose and field	Lower dose, smaller field	Higher dose, larger field (whole surgical bed)
Local control and survival	**Equivalent**	Equivalent
Other	Tumour may shrink or become better defined; thigh wounds worst	Full pathology available first

    - **Current practice favours preoperative radiotherapy** for most extremity STS, because **late toxicity is permanent whereas wound complications are temporary and treatable** (plan plastic surgical closure). Wound complications are commonest in the **lower limb, especially the thigh**.
    - Preoperative radiotherapy is particularly useful for **myxoid liposarcoma** (very radiosensitive, marked shrinkage).
- **Retroperitoneal sarcoma:** the **STRASS trial (EORTC 62092)** showed **no overall benefit of preoperative radiotherapy** on abdominal recurrence-free survival (a possible benefit in liposarcoma subgroups is debated); surgery is the main treatment (en bloc resection with adjacent organs where needed).
- **Brachytherapy** and **intensity-modulated radiotherapy (IMRT)** reduce dose to normal tissue.

Chemotherapy

- **Standard drugs: doxorubicin with or without ifosfamide**. Newer: trabectedin (myxoid/round cell liposarcoma, leiomyosarcoma), eribulin (liposarcoma), gemcitabine-docetaxel, pazopanib (non-adipocytic sarcomas), **imatinib (GIST, DFSP)**, immune checkpoint inhibitors (UPS, alveolar soft part sarcoma).
- **Adjuvant chemotherapy for adult STS is controversial**: the **EORTC 62931** trial showed **no benefit** of adjuvant doxorubicin-ifosfamide; meta-analyses (e.g. Sarcoma Meta-analysis Collaboration, Pervaiz 2008) show a **small**

benefit in local and distant recurrence with a marginal overall survival benefit. Generally offered for **high-risk (large, high-grade, deep) tumours of chemosensitive subtypes**, often as neoadjuvant therapy (ISG-STS 1001 supports histology-tailored or standard anthracycline-ifosfamide neoadjuvant regimens).

- **Chemotherapy is a standard component** for **rhabdomyosarcoma, extraskeletal Ewing sarcoma** and is commonly used in **synovial sarcoma** and **myxoid liposarcoma** (chemosensitive).
- **Isolated limb perfusion** (hyperthermic, **melphalan plus TNF-alpha**) for locally advanced, otherwise unresectable extremity sarcomas to avoid amputation.

Metastatic disease

- **Pulmonary metastasectomy** for limited, resectable lung metastases with a long disease-free interval (about 20-30% long-term survival in selected patients).
- Palliative chemotherapy, radiotherapy, targeted agents.

Follow-up

- Local recurrence: clinical examination and **MRI** (or ultrasound) of the primary site; distant: **CT chest** (or chest radiograph); every 3-4 months for 2-3 years for high-grade tumours, then 6-monthly to 5 years, then annually (most recurrences occur within 2-3 years; some subtypes recur late, e.g. synovial sarcoma, ALT).

8. Prognosis

- **Overall 5-year survival about 60-65%** for extremity STS (higher for low-grade, under 50% for large high-grade tumours).
- **Predictors of distant metastasis and death: high grade**, **large size (over 5 cm, especially over 10 cm)**, **deep location**, older age, certain subtypes (MPNST, angiosarcoma, pleomorphic liposarcoma, UPS), **local recurrence** (marker of aggressive biology).
- **Predictors of local recurrence: positive margins (the most important)**, high grade, no radiotherapy, myxofibrosarcoma/infiltrative types, previous recurrence.
- Nomograms (e.g. **Sarculator**, MSKCC) estimate individual risk.

9. Differential diagnosis of soft-tissue masses (distractors)

Lesion	Clues
Lipoma	Superficial, soft, homogeneous fat on MRI, suppresses on fat-saturation; intramuscular lipoma infiltrates muscle but is still benign
Haematoma	History of trauma or anticoagulation, should resolve; **a "haematoma" that does not resolve must be imaged with contrast MRI** (haemorrhagic sarcoma)
Myositis ossificans	After trauma; **zonal pattern: mature peripheral ossification with an immature centre** (opposite of osteosarcoma); early biopsy may mislead (pseudosarcomatous); calcification appears at 3-4 weeks
Nodular fasciitis	Rapidly growing (weeks) superficial nodule in young adults, forearm; **USP6 rearrangement**; self-limiting
Schwannoma / neurofibroma	Eccentric (schwannoma) or fusiform central (neurofibroma) along a nerve; **"split-fat", "target" and "fascicular" signs** on MRI; positive Tinel
Haemangioma / vascular malformation	Phleboliths, varies with position, serpiginous flow voids
Intramuscular myxoma	Homogeneous, very high T2, thin peripheral fat rim; with fibrous dysplasia in **Mazabraud syndrome**
Ganglion, Baker cyst	Thin-walled, peripheral enhancement only, communication with joint
Tenosynovial giant cell tumour (PVNS)	Low signal on T2 and **blooming** on gradient echo (haemosiderin); CSF1 overexpression; pexidartinib
Abscess, tuberculous cold abscess	Systemic features, rim enhancement, fluid on aspiration

Exam pearl: A soft-tissue lump that is over 5 cm, deep to fascia, enlarging, painful or recurrent needs contrast MRI and then a core needle biopsy at the sarcoma centre that will do the surgery, never an excision. Treatment is wide excision with radiotherapy (preoperative preferred: more wound complications but less permanent late toxicity, per O'Sullivan NCIC SR2); limb salvage equals amputation for survival (Rosenberg 1982).

Metastatic bone disease

1. Epidemiology

- **Metastasis is the commonest malignant lesion of bone** (far commoner than primary bone sarcoma); in a patient **over 40 with a destructive bone lesion, metastasis (or myeloma) is the most likely diagnosis**.
- **Bone is the third commonest site of metastasis** (after lung and liver).
- **Primary tumours that commonly metastasise to bone: breast, prostate, lung, kidney, thyroid**; **breast, prostate and lung account for about 80%** of skeletal metastases.
  - Breast and prostate: bone metastases in up to **70%** of advanced disease.
  - **Commonest primary in women: breast**; **in men: prostate** (lung close behind).
- **Commonest site of bone metastasis: spine** (**thoracic about 70%**, lumbar about 20%, cervical about 10%), then **pelvis, ribs, proximal femur** (the **commonest long bone**), skull and **proximal humerus**. Metastases are commonest in **red-marrow-rich axial and proximal appendicular bones**.
- **Acrometastases** (distal to elbow and knee: hand and foot) are rare; the **commonest primary is lung**.
- **Children: neuroblastoma** (commonest under 5), Wilms tumour (rare; clear cell sarcoma of kidney is the bone-seeking renal tumour), rhabdomyosarcoma, retinoblastoma.

2. Pathophysiology

- **Routes:** haematogenous (main), via **Batson's valveless vertebral venous plexus** (explains spread of **prostate and breast** cancer to the spine and pelvis when intra-abdominal or intrathoracic pressure rises), direct extension (rare).
- **"Seed and soil" hypothesis (Paget):** marrow provides a favourable microenvironment (growth factors, adhesion molecules, chemokines such as CXCL12/CXCR4).
- **The vicious cycle of osteolysis:** tumour cells secrete **PTHrP** (and IL-6, IL-11), which stimulates osteoblasts to express **RANKL** (and reduces OPG); RANKL activates **osteoclasts**; bone resorption releases **TGF-beta, IGF and calcium**

from the matrix, which stimulate further tumour growth. Basis for **bisphosphonates and denosumab** (anti-RANKL antibody).

- **Osteoblastic metastases (prostate):** tumour cells secrete **endothelin-1**, Wnt proteins and BMPs, stimulating osteoblasts; the new bone is woven and mechanically weak.
- A lesion must destroy about **30-50% of the bone mineral** before it is visible on radiographs.

Radiological pattern by primary

Pattern	Primaries
Purely lytic	**Kidney (renal cell)**, **thyroid**, lung (non-small cell, most), melanoma, GI tract, adrenal, myeloma
Osteoblastic (sclerotic)	**Prostate** (classic), carcinoid, small cell lung cancer, medulloblastoma, bladder (occasionally), Hodgkin lymphoma, treated breast cancer
Mixed	**Breast**, lung, cervix, ovary, testicular
Expansile, "blow-out" lytic	Renal and thyroid

- **Renal cell and thyroid metastases are hypervascular**: **preoperative embolisation** reduces blood loss (surgery within about 24-72 hours of embolisation); do not do an unplanned open biopsy or curettage without being ready for major bleeding.

3. Clinical features

- **Pain**: the commonest symptom; initially **night and rest pain** (biological), later **mechanical/functional pain on weight bearing** (impending fracture).
- **Pathological fracture**: first presentation in some patients.
- **Hypercalcaemia**: the **commonest metabolic complication** (up to 10-30% of patients with advanced cancer; breast, lung, myeloma; renal through PTHrP): thirst, polyuria, confusion, constipation, nausea, arrhythmias. Treat with **IV fluids, IV bisphosphonate (zoledronic acid)**, **denosumab** if refractory or renal impairment, calcitonin for rapid short-term effect, steroids in lymphoma/myeloma.

- **Metastatic spinal cord compression (MSCC)**: back pain (often thoracic, band-like, worse lying down or with straining) precedes neurological deficit; weakness, sensory level, sphincter disturbance late.
- Marrow failure (anaemia), constitutional symptoms.
- **Skeletal-related events (SREs)**: pathological fracture, spinal cord compression, need for radiotherapy or surgery to bone, hypercalcaemia.

4. Evaluation of a destructive bone lesion in a patient over 40

Rougraff diagnostic strategy (Rougraff et al., JBJS 1993): patient with a skeletal metastasis of unknown origin

1. **History** (smoking, weight loss, previous cancer, haematuria, breast lump) and **examination** (breast, prostate, thyroid, abdomen, lymph nodes, skin).
2. **Laboratory:** FBC, ESR, **calcium**, **alkaline phosphatase**, renal and liver function, **serum and urine protein electrophoresis/free light chains (myeloma)**, **PSA** (men), thyroid function and others as indicated. Tumour markers (CEA, CA 15-3, CA-125) are of limited value.
3. **Plain radiographs** of the affected bone (whole bone) and **chest radiograph**.
4. **CT of chest, abdomen and pelvis**.
5. **Whole-body bone imaging: technetium-99m bone scan** (or PET-CT) to find other lesions and the most accessible site for biopsy.
6. **Biopsy** of the most accessible lesion (core needle, image-guided).
7. This approach **identified the primary in about 85%** of cases. **Lung (non-small cell) carcinoma was the commonest unknown primary**, followed by **kidney**. In about 10-15% no primary is found ("carcinoma of unknown primary").

> Exam pearl: Never fix a solitary destructive bone lesion until it is proven to be a metastasis. A "solitary metastasis" may be a primary bone sarcoma (needs wide resection, and intramedullary nailing would contaminate the whole bone), myeloma or renal carcinoma (needs embolisation). Biopsy first, following sarcoma biopsy principles; even with a known primary cancer, a first and solitary bone lesion should be biopsied.

Imaging details

- **Bone scintigraphy (Tc-99m MDP)**: sensitive whole-skeleton survey; **false negatives in purely lytic lesions** (**myeloma**, some **renal and thyroid** metastases, aggressive lytic lesions) and LCH; "**superscan**" (diffuse uptake with absent renal activity) in widespread prostate/breast metastases.
- **PET-CT**: FDG-avid lytic metastases; **PSMA PET-CT** for prostate cancer staging.
- **MRI**: **the most sensitive for marrow and spinal disease**, epidural extension and cord compression; whole-spine MRI for suspected MSCC.
- **CT**: cortical destruction, fracture risk, surgical planning; CT-guided biopsy.

Spinal clues

- **Vertebral body is involved first**; the **pedicle is destroyed on the AP radiograph ("winking owl" sign)**, because the body and posterior elements are involved together. Myeloma often spares the pedicle early.
- **Disc space is preserved** (tumours rarely cross the disc), in contrast to **pyogenic infection**, which destroys the disc and both endplates; **TB** may spare the disc early and spread subligamentously.
- **Malignant versus osteoporotic vertebral collapse on MRI:**

Feature	Malignant	Osteoporotic
Marrow signal	Diffuse T1 low signal replacing the **whole body**, often **multiple non-contiguous levels**	Band-like oedema, **preserved normal fatty marrow** in part of the body
Posterior cortex	**Convex bulge**	Retropulsed bony fragment, angular
Pedicle / posterior elements	**Involved**	Spared
Epidural/paravertebral mass	Present	Absent or thin
Other	Diffusion restriction, enhancement	**Fluid sign**, intravertebral vacuum cleft

5. Predicting pathological fracture

Mirels scoring system (Mirels 1989), long bones

Variable	1 point	2 points	3 points
Site	Upper limb	Lower limb	**Peritrochanteric**
Pain	Mild	Moderate	**Functional (mechanical)**
Lesion type	Blastic	Mixed	**Lytic**
Size (fraction of bone diameter on radiograph)	Under one-third	One-third to two-thirds	**Over two-thirds**

Total score	Fracture risk	Recommendation
7 or less	About 4% (low)	**Radiotherapy and observation**
8	About 15%	**Clinical judgement**
9 or more	**About 33% or more**	**Prophylactic fixation**

- Mirels score has **high sensitivity but poor specificity** (many patients scoring 9 would not fracture) and is less reliable for the femur at the margins of the scale; **functional pain is the most important clinical variable**.
- **Other criteria for prophylactic fixation of the femur:**
  - **Harrington's criteria: cortical destruction of 50% or more**, **lesion of 2.5 cm or more in the proximal femur**, **pathological avulsion of the lesser trochanter**, and **persistent pain on weight bearing despite radiotherapy**.
  - **Axial cortical involvement over 30 mm** (or circumferential involvement over 50%) on radiographs: the predictor validated in the **Dutch Femoral Metastases study (Van der Linden et al., 2004)**; lesions below this can be treated with radiotherapy alone.
  - **CT-based structural rigidity analysis (CTRA)** and **finite element models** are more accurate but not yet widely available.
- **Expected survival** guides the decision: **orthopaedic surgery is generally considered when expected survival is more than about 6 weeks** (long

bones) and **3 months** (spine) in classic teaching; tools such as **PATHFx**, **SPRING** and **Katagiri** scores estimate survival.

- **Healing of pathological fractures (Gainor and Buchert 1983): myeloma highest (about 65-70%)**, **renal about 45%**, **breast about 35-40%**, **lung essentially 0%**. Healing depends on survival over 6 months; radiotherapy does not prevent healing but chemotherapy and poor nutrition reduce it. **Implants must therefore be designed to last without bone healing.**

6. Principles of surgical management (long bones)

- **Aims:** pain relief, **immediate weight bearing and function**, nursing care, and a construct that **outlasts the patient** (single operation).
- **Multidisciplinary discussion** (oncologist, radiotherapist, orthopaedic oncologist; BOA guidance on metastatic bone disease recommends this for all impending and actual fractures), optimise calcium, haemoglobin, nutrition; **embolise hypervascular lesions** (renal, thyroid, some myeloma).
- **Image the whole bone** (and the other femur) before fixation to detect additional lesions that must be protected.
- **Principles of fixation:**
  - **Protect the whole bone**: e.g. **long cephalomedullary (reconstruction) nail** for the femur so the femoral neck and entire shaft are protected.
  - Use **load-bearing or load-sharing** constructs that allow **immediate full weight bearing**.
  - **Curettage of the lesion and cement augmentation (PMMA)** to fill defects and improve screw purchase (with plates or nails).
  - **Replace rather than fix** where periarticular bone is destroyed: fractures of the **femoral head and neck do not heal** reliably: use **cemented (long-stem) hemiarthroplasty or total hip arthroplasty**, not internal fixation.
  - **Endoprosthetic (tumour/megaprosthesis) replacement** for extensive metaphyseal or periarticular destruction, failed fixation, or **solitary renal/thyroid metastasis treated for cure**.
- **Postoperative radiotherapy** to the **whole operative field** (entire implant length), usually started **about 2 weeks after surgery** once the wound is healing: **Townsend et al. (1994)** showed **fewer second procedures and better function** with postoperative radiotherapy.

- **Cardiopulmonary complications: fat and tumour embolism** with intramedullary reaming and **cemented long stems** (bone cement implantation syndrome): vent the canal, lavage, careful cementing, communicate with anaesthesia.

Choice of procedure by site

Site	Preferred treatment
Femoral head/neck	**Cemented hemiarthroplasty or THA** (long stem if more distal lesions); acetabulum assessed
Intertrochanteric / subtrochanteric	**Long cephalomedullary nail** (plus cement); **proximal femoral replacement** if extensive destruction or failed fixation
Femoral shaft	**Locked intramedullary nail** (protect the neck with a reconstruction nail)
Distal femur	Locking plate plus cement, retrograde nail, or **distal femoral replacement**
Proximal humerus	Intramedullary nail, plate plus cement, or **proximal humeral endoprosthesis/hemiarthroplasty**
Humeral shaft	**Locked intramedullary nail** (or plate plus cement)
Tibia	Intramedullary nail; distal lesions plate plus cement
Pelvis (non-weight-bearing ilium, pubic rami)	Radiotherapy, non-operative

Periacetabular metastases: Harrington classification

Class	Defect	Treatment
I	Lateral cortices, superior and medial walls **intact**	**Conventional cemented THA**
II	**Medial wall deficient**	THA with a **protrusio cage/ring or mesh** to transfer load to the rim

Class	Defect	Treatment
III	**Lateral cortex and superior dome (roof) deficient**	**Harrington procedure: Steinmann pins (or screws) driven from the iliac crest into the acetabular region**, reinforced with **cement**, plus cage and cemented cup
IV	**Solitary lesion treated for cure** (or pelvic discontinuity)	**En bloc resection** and reconstruction

- Newer minimally invasive options: **percutaneous acetabuloplasty** (cementoplasty), ablation with screw fixation and cement.

7. Metastatic spine disease

Assessment framework: NOMS (Memorial Sloan Kettering)

- **N - Neurological:** degree of **epidural spinal cord compression (ESCC, Bilsky grade)** and myelopathy/radiculopathy.
- **O - Oncological: radiosensitivity** of the tumour.
- **M - Mechanical:** instability (**SINS**).
- **S - Systemic:** extent of disease, comorbidity, ability to tolerate surgery, expected survival.

Bilsky (ESCC) grading on axial T2 MRI

Grade	Description
0	Bone-only disease
1a	Epidural impingement **without deformation of the thecal sac**
1b	Thecal sac deformed, **no spinal cord abutment**
1c	Thecal sac deformed, **spinal cord abutted but not compressed**
2	**Cord compression, CSF still visible** around the cord
3	**Cord compression, no CSF visible**

- **Grades 0-1 = low grade**: radiotherapy (or SBRT); **grades 2-3 = high grade**: decompression is considered for radioresistant tumours.

Radiosensitivity

Radiosensitive	Radioresistant (to conventional external-beam radiotherapy)
Lymphoma, myeloma/plasmacytoma, seminoma, small cell lung, **breast**, **prostate** (most), Ewing	**Renal cell, thyroid, melanoma, sarcoma**, non-small cell lung, colon, hepatocellular

- **Stereotactic body radiotherapy (SBRT)** delivers high doses accurately and overcomes radioresistance; "**separation surgery**" (posterolateral decompression creating a 2-3 mm gap between tumour and cord, plus stabilisation) followed by SBRT is the MSKCC approach for high-grade ESCC of radioresistant tumours.

Spinal Instability Neoplastic Score (SINS; Spine Oncology Study Group, 2010)

Component	Scores
Location	**Junctional** (occiput-C2, C7-T2, T11-L1, L5-S1) **3**; mobile (C3-C6, L2-L4) 2; semi-rigid (T3-T10) 1; rigid (S2-S5) 0
Pain	**Mechanical pain 3**; occasional non-mechanical pain 1; pain-free 0
Bone lesion	Lytic 2; mixed 1; blastic 0
Radiographic alignment	**Subluxation/translation 4**; de novo deformity (kyphosis/scoliosis) 2; normal 0
Vertebral body collapse	Over 50% collapse 3; under 50% collapse 2; no collapse with over 50% body involved 1; none of these 0
Posterolateral involvement (facet, pedicle, costovertebral joint)	Bilateral 3; unilateral 1; none 0

- **Total 0-6 = stable**; **7-12 = indeterminate (potentially unstable)**; **13-18 = unstable**. **Surgical consultation is recommended for a score of 7 or more.**

Survival scores used to choose surgery

Score	Components	Interpretation
Revised Tokuhashi (0-15)	General condition (Karnofsky), extraspinal bone metastases, vertebral metastases, visceral metastases, primary site, neurological deficit	**0-8: under 6 months** (conservative/palliative); **9-11: 6 months or more** (palliative surgery); **12-15: 1 year or more** (excisional surgery)
Tomita (2-10)	Primary tumour growth rate (slow 1, moderate 2, rapid 4), visceral metastases (none 0, treatable 2, untreatable 4), bone metastases (solitary 1, multiple 2)	**2-3: wide or marginal excision (en bloc, long-term control)**; **4-5: marginal or intralesional excision**; **6-7: palliative surgery**; **8-10: supportive care**

Treatment of metastatic spinal cord compression

- **Immediate: dexamethasone** (e.g. 16 mg/day; higher bolus doses debated), analgesia, flat bed rest and spinal precautions if instability is suspected; **urgent whole-spine MRI (UK NICE: within 24 hours)**; spinal oncology and clinical oncology referral.
- **Patchell et al. (Lancet 2005) randomised trial: direct decompressive surgery (circumferential decompression and stabilisation) plus radiotherapy** versus **radiotherapy alone** for MSCC from a solid tumour:
  - **84% vs 57% able to walk after treatment**; **non-ambulant patients regained walking in 62% vs 19%**; walking retained longer (median 122 vs 13 days); less steroid and opioid use; trial stopped early for benefit.
  - **Inclusion:** single area of compression, **expected survival 3 months or more**, **paraplegia for less than 48 hours**, **not radiosensitive tumours** (lymphoma, myeloma, leukaemia, germ-cell excluded).
- **Radiotherapy alone:** radiosensitive tumours, poor surgical candidates, multilevel disease, long-standing complete paraplegia (over 48 hours), short life expectancy.
- **Surgical options:** posterior decompression and **instrumented stabilisation** (laminectomy alone is contraindicated when the anterior column is involved, as it worsens instability), separation surgery, corpectomy and reconstruction, **en bloc spondylectomy (Tomita)** for solitary renal or

thyroid metastasis with good prognosis; **minimally invasive percutaneous screw fixation**.

- **Vertebral augmentation (vertebroplasty or kyphoplasty)** for painful pathological compression fractures without significant epidural disease or instability: **CAFE trial (Berenson et al., Lancet Oncology 2011)**: **balloon kyphoplasty gave rapid and sustained improvement in pain and function** compared with non-surgical care. Cement leakage is the commonest complication.

8. Non-surgical treatment

Radiotherapy

- Pain relief in about **60-70%** of patients, often within 2-4 weeks.
- **Single fraction 8 Gy gives the same pain relief as multiple fractions** (e.g. 30 Gy in 10, 20 Gy in 5), confirmed by **RTOG 9714** and the **Dutch Bone Metastasis Study**; **retreatment rate is higher after single fraction (about 18-20% vs 8-10%)**. Single fraction is preferred for uncomplicated painful metastases and poor prognosis.
- **Stereotactic (SBRT)** for oligometastases, radioresistant tumours and re-irradiation; risk of **vertebral compression fracture** after SBRT.
- **Radiopharmaceuticals: radium-223** (alpha emitter; **ALSYMPCA trial**: improved overall survival in castration-resistant prostate cancer with bone metastases), strontium-89 and samarium-153 (palliation, marrow suppression).

Bone-targeted agents

- **Bisphosphonates (zoledronic acid):** inhibit osteoclasts (**farnesyl pyrophosphate synthase** in the mevalonate pathway); **reduce skeletal-related events** in breast, prostate, lung cancer and myeloma; treat hypercalcaemia.
- **Denosumab (monoclonal antibody against RANKL): superior to zoledronic acid in delaying the first SRE** in breast (Stopeck et al., 2010) and prostate cancer (Fizazi et al., 2011); no renal dose adjustment but risk of **hypocalcaemia** (give calcium and vitamin D); rebound bone loss on stopping.

- **Complications: osteonecrosis of the jaw** (dental assessment before starting), **atypical femoral fractures** (lateral cortical beaking: watch for thigh pain), acute-phase reaction, renal toxicity (zoledronic acid).
- **Systemic oncological therapy** (hormonal therapy for breast and prostate, targeted agents, immunotherapy) controls disease and improves healing.
- **Analgesia** (WHO ladder), physiotherapy, orthoses, palliative care.

9. Special situations

- **Solitary renal cell carcinoma metastasis: wide (en bloc) excision** may **improve survival** (especially metachronous solitary lesions after nephrectomy with a long disease-free interval); **embolise preoperatively**; renal metastases respond poorly to radiotherapy and fixation alone often fails (progression around the implant), so **resection and endoprosthetic reconstruction** is favoured.
- **Thyroid metastases:** radioiodine-avid lesions may respond to **radioactive iodine**; hypervascular; solitary lesions resected.
- **Prostate:** blastic lesions are painful but fracture less often; respond to androgen deprivation; PSMA-targeted therapy (lutetium-177 PSMA).
- **Breast:** mixed lesions, hormone-responsive; long survival possible, so durable reconstructions are justified.
- **Lung:** shortest survival (often months), fractures rarely heal: simplest effective procedure.
- **Impending fracture versus completed fracture: prophylactic fixation** of an impending fracture gives **shorter hospital stay, less blood loss, faster return to walking and better survival** than fixation after the fracture (retrospective data); hence treat impending fractures early.

10. Differential diagnosis of a lytic lesion in an older adult (distractors)

Condition	Clue
Myeloma	Punched-out lesions, normal ALP, cold bone scan, M-protein, anaemia, renal failure

Condition	Clue
Primary bone sarcoma (chondrosarcoma, UPS of bone, osteosarcoma, secondary sarcoma in Paget disease)	Solitary lesion, matrix calcification, periosteal reaction, soft-tissue mass; **biopsy before fixation**
Lymphoma	Permeative, large soft-tissue mass, well patient
Brown tumour of hyperparathyroidism	High calcium, high PTH, low phosphate, subperiosteal resorption of phalanges
Paget disease	Coarse trabeculae, cortical thickening, enlarged bone, raised ALP; **osteoporosis circumscripta** of skull in lytic phase
Osteoporotic insufficiency fracture	Typical sites (sacrum H-sign, femoral neck), MRI signal pattern
Infection (pyogenic, tuberculosis)	Disc involvement in spine, fever, raised CRP
Giant cell tumour	Epiphyseal-metaphyseal, eccentric, skeletally mature (20-40 years)

Exam pearl: Breast, prostate, lung, kidney and thyroid cause most bone metastases; spine is the commonest site and proximal femur the commonest long bone. Mirels 9 or more means prophylactic fixation. Fix the whole bone, use cement, replace (do not fix) the femoral neck, give postoperative radiotherapy to the whole implant, and embolise renal and thyroid metastases. For spinal cord compression from a radioresistant solid tumour, decompression with stabilisation plus radiotherapy beats radiotherapy alone (Patchell 2005).